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Possibilistic Variational Inference

Master's Internship Report
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*Throughout this report, section-level references are spelled out in full (*Section X*), while subsection-level references use the symbol § (*§ X.Y*)*

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Abstract

Possibilistic variational inference (VI) replaces probability measures with supremum-normalised possibility functions to capture epistemic uncertainty without imposing additive measures on deterministic parameters. Singh et al. [1] established a maxitive Donsker–Varadhan formula, yielding dual lower and upper Consistency Bound Objectives (CBOs) and the `CBOpt` second-order optimisers. In practice, however, these methods restrict candidates to isotropic Gaussians whose scalar precision ϑ is fixed or adjusted via heuristic curvature tracking, treating dispersion as an external hyperparameter rather than deriving it variationally.

This report resolves that gap. After reproducing a lighter variant of the reference LeNet-5/Fashion-MNIST benchmark of [1] within two standard deviations and showing on four tractable targets that misspecification stems from geometric mismatch rather than divergence flaws, it develops a deductive progression for learning dispersion across four matrix geometries: isotropic (iso), diagonal (diag), low-rank-plus-diagonal (LRD), and layer-wise block-diagonal (BD). To prevent unpenalised CBO collapse, this report introduces an extremal-volume penalty with scale $\zeta = d/2$. For isotropic and diagonal precisions, stationarity yields implicit fixed points solved via damped Picard iterations with anti-collapse and positivity safeguards. For LRD and BD matrices, the singular rank-one target is regularised, with positive definiteness proving sign-asymmetric: unconditional for $s = +1$, but feasibility-constrained for $s = -1$. Extending derivations by Shelvia Wongso connecting the lower CBO to Sharpness-Aware Minimization (SAM), we analyse the resulting Gaussian relaxation to extract an adaptive perturbation radius directly from the learned spread.

Linear per-step inversion algorithms are evaluated on a quadratic toy problem and LeNet-5/Fashion-MNIST. On the toy, experiments confirm the complexity hierarchy LRD < BD < iso < diag and high sensitivity to the penalty scale, justifying $\zeta = d/2$. On LeNet-5, `uCBOpt-diag` and `uCBOpt-BD` lower expected calibration error (ECE) to 0.020 and 0.014 (against 0.025 for fixed `uCBOpt` and 0.019 for 64-sample IVON) with single deterministic passes, improving accuracy and negative log-likelihood (NLL). In the SAM setting, the first-order linearisation gap scales empirically as $\mathcal{O}(\rho^2)$, while `lCBOpt-iso` exhibits numerical insensitivity across tolerance sweeps. Yet, diagnostics reveal clear operational limits: Picard contraction bounds fail at network scale ($L_\Phi \sim 10^7$–10^8), and raw iterations satisfy $H_s \succ 0$ on 0.0% of steps, restricting precision to the boundary cap (relegating SAM sensitivity tests to suboptimal regimes). Finally, the global $s = -1$ safeguard in `lCBOpt-LRD` triggers 100% sign flips, an obstruction mitigated under `lCBOpt-BD`, which preserves $s = -1$ feasibility on 65.5% of active blocks.

Keywords. Possibility theory, variational inference, epistemic uncertainty, maxitive Donsker–Varadhan, sharpness-aware minimization, second-order optimisation, learned precision.

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1 Introduction

1.1 Context of the internship and relation to the MVA curriculum

This internship was carried out between April and August 2026 under the joint supervision of Badr-Eddine Chérief-Abdellatif and Jérémie Houssineau, first at the Laboratoire de Probabilités, Statistique et Modélisation (LPSM, CNRS and Sorbonne Université) in Paris, then at the School of Physical and Mathematical Sciences (SPMS, Nanyang Technological University) in Singapore. The two supervisors are, with Jasraj Singh and Shelvia Wongso, the authors of the paper [1] that this work extends. The internship began by acquiring the background in Bayesian and variational inference, possibility theory, and information theory required for [1] (summarised in Section 3), before moving to the reproduction and diagnostic work of Sections 4 and 5 and the extension developed in Sections 6 and 7.

This work builds on several courses from the master's program. *Bayesian Machine Learning* and *Computational Statistics* provide the variational-inference machinery that the possibilistic construction mirrors and the convergence analysis of the iterative approximation schemes. *Convex Optimization and Applications in Machine Learning* supplies the convex-analysis and duality background behind possibilistic exponential families and the dual variational objectives used throughout, while *Deep Learning* provides the experimental setting.

1.2 Is probability the right language for epistemic uncertainty?

Probability theory is the standard language for uncertainty, and for aleatoric uncertainty (the intrinsic randomness of a data-generating process) it is the right one. The difficulty arises with epistemic uncertainty, a lack of knowledge about a fixed, deterministic quantity $\theta^\star \in \Theta$, with no repeatable experiment from which it is drawn. Forcing a probability measure onto such a quantity conflates the two sources of uncertainty, and the additivity constraint $\mathbb{P}(A) + \mathbb{P}(A^c) = 1$ makes complete ignorance inexpressible, since one cannot assign full credibility to both an event and its complement.

Whether this matters in practice has been debated extensively, and this report does not try to settle that debate. We point to two discussions from this literature that motivate the possibilistic route explored below.

The first is a paradox concerning the improper prior: on $\Theta = (0, \infty)$ with no information, $\pi(\theta) \propto 1$ is not integrable, and improper priors can yield improper posteriors. For example, with $X \mid \theta \sim \mathrm{Unif}(0, \theta)$ and $\pi(\theta) \propto 1$, $\pi(\theta \mid x) \propto \theta^{-1} \mathbf{1}_{\theta \geq x}$, whose normalising integral $\int_x^\infty \theta^{-1} \, \mathrm{d}\theta$ diverges. Even when the posterior is proper, marginalisation can produce inconsistent inferences, as in the marginalisation paradoxes of Dawid, Stone, and Zidek [2]: improper priors do not define probability measures, so standard conditioning and marginalisation may fail. Jeffreys' prior, $\pi_J \propto \sqrt{\det I(\theta)}$, is invariant under reparametrisation but can remain improper on unbounded Θ: for $X \mid \mu \sim \mathcal{N}(\mu, \sigma^2)$ with known σ^2, $I(\mu) = 1/\sigma^2$, so $\pi_J(\mu) \propto 1$ and $\int_{\mathbb{R}} \pi_J(\mu) \, \mathrm{d}\mu = \infty$. And since the Fisher information depends on the likelihood, Jeffreys' prior makes the representation of ignorance depend on the data-generating model, undesirable when the prior is meant to represent ignorance independently of those assumptions.

The second concerns probability dilution, and is more contested. Let $A \subset \Theta$ be a bounded set whose probability we wish to assess, such as the collision region in satellite conjunction analysis, and model epistemic uncertainty about the fixed but unknown $\theta^\star$ by $\theta^\star \sim \mathcal{N}(\mu, \sigma^2)$. As $\sigma \to \infty$, $\mathbb{P}(\theta^\star \in A) \to 0$: less precise trajectory information yields a smaller computed collision probability, even though the physical trajectory has not become safer. Balch, Martin, and Ferson [3] argue this can be misleading, terming it a symptom of false confidence: for certain false propositions $A \not\ni \theta^\star$, probabilistic inference can assign A an arbitrarily high degree of belief with arbitrarily high probability under the true data-generating distribution. This reading is not shared by every Bayesian: for many, a posterior probability is not meant to certify coverage in the sense the false-confidence critique requires, a broader tension between Bayesian and frequentist readings of a posterior that runs through the discussion around Fraser's [4] comparison of Bayes posteriors and confidence statements.

Possibility theory, introduced by Zadeh [5] and developed by Dubois and Prade [6], replaces additivity by maxitivity, $\overline{P}(A \cup B) = \max(\overline{P}(A), \overline{P}(B))$. Following Houssineau [7], a possibility function is any $f : \Theta \to [0, 1]$ with $\sup_{\theta \in \Theta} f(\theta) = 1$, with associated possibility measure (or maxitive outer measure) $\overline{P}(A) = \sup_{\theta \in A} f(\theta)$. Three consequences follow directly. First, $\overline{P}(A) = \overline{P}(A^c) = 1$ is admissible, expressing total absence of information. Second, $\mathbf{1}_\Theta$ is a proper non-informative prior, with no integrability condition, and invariant under bijective reparametrisation since the supremum pushforward of a constant remains constant. Third, marginalisation reduces to taking the supremum rather than integrating, resolving the classical marginalisation paradoxes from improper priors.

Figure 1 shows how the possibilistic picture avoids the dilution above, for a fixed target $A = [0.4, 0.6]$ and true value $\theta^\star = 0.5$. On the left, $\mu = 0.5 \in A$: the probabilistic risk $\mathbb{P}(\theta \in A)$ (blue) decays to 0 as σ grows, even though μ never leaves A, while the possibilistic credibility $\overline{P}(\theta \in A)$ (orange) stays at 1 for every σ, since the mode of f lies in A. On the right, $\mu = 0.3 \notin A$: both curves start low, but as $\sigma \to \infty$ they diverge, $\mathbb{P}(\theta \in A) \to 0$ while $\overline{P}(\theta \in A) \to 1$, since a sufficiently uninformative possibility function assigns high credibility to every bounded set regardless of where its mode sits. The probabilistic curve tracks only σ; the possibilistic one tracks whether $\mu \in A$.

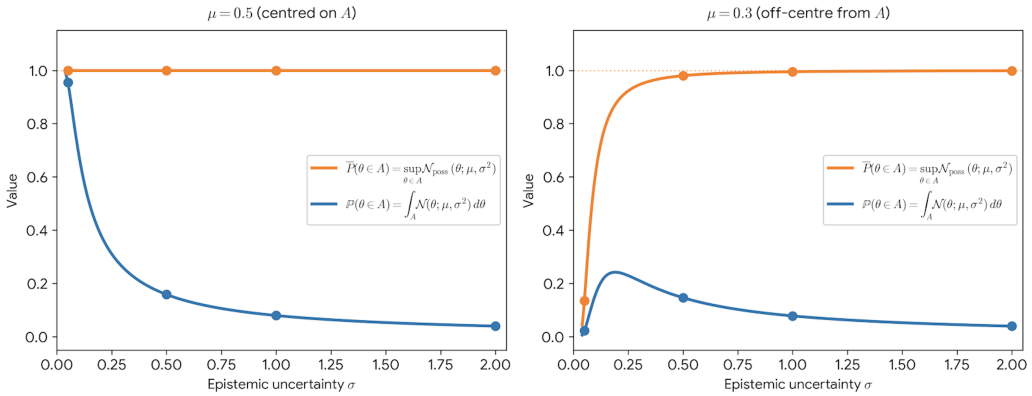


Figure 1: Probability dilution, with $A = [0.4, 0.6]$ and $\theta^\star = 0.5$. Blue is the probabilistic risk $\mathbb{P}(\theta \in A)$, orange the possibilistic credibility $\overline{P}(\theta \in A)$. Left, $\mu = 0.5 \in A$, so $\overline{P}(\theta \in A) = 1$ for all σ while $\mathbb{P}(\theta \in A) \to 0$. Right, $\mu = 0.3 \notin A$, so $\overline{P}(\theta \in A) \to 1$ as $\sigma \to \infty$.

1.3 From probabilistic to possibilistic variational inference

Probabilistic variational inference (VI) (recalled in Section 3) approximates an intractable target with a tractable candidate by optimisation. Two motivations justify recasting this in the possibilistic setting. The first is practical: no sampling is needed at inference, as a single deterministic forward pass at the candidate mode suffices, unlike IVON, MC-Dropout, SWAG, or Laplace, which all require several samples per prediction to approximate posterior expectations. The second is structural: probabilistic VI replaces an intractable integration, the marginal likelihood or evidence, by a tractable optimisation problem. In the possibilistic framework, the target normalising constant $Z_{\max} = \sup_\theta e^{-\hat{\ell}(\theta)}$ is already formulated as an optimisation. The variational reformulation is therefore optimisation-to-optimisation rather than integration-to-optimisation, trading a supremum over Θ for one over a tractable candidate family. This makes the derivations particularly natural in the dual lower/upper characterisation introduced next (where the evidence lower bound, or ELBO, provides only a one-sided bound) and underlies the direct connection to Sharpness-Aware Minimization (SAM) developed in Section 7.1.

Singh et al. [1] transpose VI to this setting on the foundations of possibility theory. Writing $\hat{\ell} = \ell - \log f_0$ for the regularised loss under a prior possibility function f_0, the probabilistic evidence $Z_{\text{add}} = \int_\Theta e^{-\hat{\ell}(\theta)}\, \mathrm{d}\theta$ is replaced by the maxitive marginal likelihood $Z_{\max} = \sup_{\theta \in \Theta} e^{-\hat{\ell}(\theta)}$, and the exact possibilistic posterior is $g^\star_{\max} = e^{-\hat{\ell}}/Z_{\max}$. The central theoretical tool is a maxitive analogue of the Donsker–Varadhan formula, which yields two dual variational characterisations: a lower objective whose maximisers are candidates bounded pointwise above by $g^\star_{\max}$, and an upper objective whose minimisers are candidates bounded pointwise below by $g^\star_{\max}$. The posterior is the unique candidate satisfying both bounds simultaneously.

Building upon this theoretical foundation, Singh et al. [1] derive a signed Newton-like update rule that gives rise to the `CBOpt` family of optimisers. On standard image classification benchmarks, these algorithms demonstrate empirical performance competitive with variational Bayesian baselines such as IVON [8], without requiring Monte-Carlo sampling at inference time. In practice, however, the variational candidate is restricted to an isotropic Gaussian possibility function $\mathcal{N}^*(\theta; \eta, \vartheta^{-1} I)$, in which the scalar precision parameter ϑ is either fixed by hand or adjusted during training through a heuristic curvature-tracking envelope. More generally, the candidate could incorporate a full positive-definite precision matrix $S = \Sigma^{-1}$ (the notation used in Section 6). Yet this anisotropic freedom is never leveraged, and in neither case is the precision derived or optimised from the variational objective itself. Resolving this limitation constitutes the main objective of the present work.

1.4 Personal Contributions

The contributions developed in this report follow directly from the four main sections, with corresponding **repository links** for contributions 1, 2, and 4 provided in Appendix A: reproducing the reference baseline, investigating candidate misspecification across tractable landscapes, developing the precision learning framework, and evaluating the resulting optimisers and the connection to robust optimization.

1. **Reproducing [1]: `uCBOpt` at fixed precision** (Section 4). I reproduce a lighter variant of the LeNet-5/Fashion-MNIST benchmark of [1], matching all five metrics against the reference within two standard deviations and recovering the qualitative ranking IVON $\approx$ `uCBOpt` $>$ AdamW (the ranking variations on out-of-distribution (OOD) detection remaining within baseline variance), thereby validating the experimental setup on which subsequent contributions rely.

2. **Landscapes on tractable posteriors** (Section 5). On four tractable target distributions, I evaluate candidate misspecification under both variational objectives ($s = \pm 1$). The analysis shows that gaps between variational candidates and true posteriors come from geometric support, tail, or multimodality mismatch rather than the variational divergence itself, the mismatched gaps remaining invariant under order-p softening of $D_{\max}$.

3. **Constructing a learned precision** (Section 6). Moving beyond heuristics where precision is fixed or adjusted by hand, I formulate the learning of candidate spread through a deductive progression across four structures: isotropic (iso), diagonal (diag), low-rank-plus-diagonal (LRD), and layer-wise block-diagonal (BD) (detailed in §8.1, **A chain of results**). To resolve the collapse of unpenalised Consistency Bound Objectives (CBOs) (Lemma 6.3), I introduce an extremal-volume penalty (Theorem 6.5). For isotropic and diagonal precisions, the resulting stationarity conditions define implicit fixed points solved via damped Picard iterations with anti-collapse and positivity caps (Proposition 6.10, Theorem 6.14). For full and block-diagonal matrices, the singular rank-one target (Proposition 6.18) is regularised into a LRD form with sign-asymmetric positive-definiteness guarantees (Theorems 6.20–6.24).

4. **Applying and validating the construction** (Section 7). Following derivations by Shelvia Wongso [9] connecting the lower CBO to SAM, I investigate the induced adaptive perturbation radius (§7.1). I formulate the practical optimisers with linear per-step complexity (Proposition 7.3, §7.2) and evaluate them across an exact quadratic toy and LeNet-5/Fashion-MNIST (§7.3). On the toy, experiments reveal high sensitivity to the penalty scale, justifying fixing $\zeta = d/2$, and validate the complexity hierarchy LRD $<$ BD $<$ iso $<$ diag while highlighting GPU launch overheads. On LeNet-5, `uCBOpt-diag` and `uCBOpt-BD` improve expected calibration error (ECE) over fixed baselines and multi-sample IVON from a single deterministic pass, alongside gains in accuracy and negative log-likelihood (NLL). Diagnostics reveal, however, that raw Picard iterations satisfy $H_s \succ 0$ on only 0.0% of steps, with the operational precision tracking the cap boundary rather than an interior optimum, while the global $s = -1$ safeguard in `lCBOpt-LRD` requires 100% sign flips, partially restored to 65.5% active coverage by `lCBOpt-BD`.

2 Notation

Table 1 summarises the notation used throughout the report. The first block (Section 3) is inherited from the reference paper [1], the second block (Section 6 and 7) introduces my own notation.

Table 1: Notation used in the report.

Symbol	Role	First defined
Section 3		
Θ	Parameter space	§3.3
π	Prior (density, or possibility function in the possibilistic case)	§3.3
$q^\star_{\text{add}}$	Additive (Bayesian) posterior	§3.3
Z_{add}	Additive evidence, $\int e^{-\ell}\pi$	§3.3
$\mathrm{KL}(q\|p)$	Kullback–Leibler divergence	§3.3
$\mathrm{ELBO}(q)$	Evidence lower bound	§3.3
$\mathcal{F}(\Theta)$	Set of possibility functions on Θ	§3.3
$\bar{P}$	Outer probability measure	§3.3
$\preceq$	Specificity order on $\mathcal{F}(\Theta)$	§3.3
$\overline{\mathbb{E}}^*[\theta]_f$	Possibilistic expectation (mode) of f	§3.3
$\bar{I}_f$	Possibilistic precision of f	§3.3
$\mathcal{N}^*(\theta;\mu,\Sigma)$	Normal possibility function	§3.3
$g^\star_{\max}$	Possibilistic posterior	§3.3
$Z_{\max}$	Maxitive marginal likelihood, $\sup_\theta e^{-\hat{\ell}(\theta)}$	§3.3
$\hat{\ell}$	Regularised loss, $\ell - \log\pi$	§3.3
$\underline{\mathrm{CBO}}$, $\overline{\mathrm{CBO}}$	Lower / upper consistency bound objective	Eq. (3)
$\overline{D}_{\max}(g\|f)$	Max-relative entropy	Eq. (4)
g_λ	Exponential-family possibility function	§3.4
λ	Natural parameter of the exponential family	§3.4
A, $A^\dagger$	Log-partition function and its Legendre transform	§3.4
$s \in \{\pm 1\}$	Sign of the CBO: $+1$ upper, -1 lower	§3.4
$\mathrm{CBO}_s(g)$	Signed consistency bound objective, unified over s	§3.4
η	Mode of the candidate	§3.4
ϑ, $\boldsymbol{\vartheta}$	Isotropic / diagonal candidate precision	§3.4
δ	Prior precision (L^2/weight-decay coefficient)	§3.4
H_s	Signed curvature, $s(\nabla^2\hat{\ell} - \nabla^2 A_0)$	§3.4
ρ_t	Learning rate (not to be confused with SAM radius ρ)	Eq. (6)
h_t, g_t, $\bar{g}_t$	Curvature / gradient / bias-corrected gradient (EMA)	§3.4
$\beta_1, \beta_2, \beta_3$	EMA coefficients	§3.4
Section 6 and 7		
H^X_s, θ^X_s, u^X_s	Signed curvature / inner extremiser / Newton step under precision structure $X \in \{\emptyset, \mathrm{diag}, \mathrm{LRD}, \mathrm{BD}\}$	§6
S	General precision matrix, $S \in \mathcal{S}^d_{++}$	§6.1
$\mathrm{vol}_\alpha(S)$	Volume of the level set $\{g_{\eta,S} \geq \alpha\}$	Def. 6.4
μ	Lagrange multiplier for the extremal-volume selection	Thm. 6.5
ζ	Coefficient of the log-volume penalty	Prop. 6.9
Φ, Φ^{diag}_j	Picard map, isotropic / diagonal	Eq. (13)
w	Picard damping weight, $w \in (0,1]$	Eq. (13)
T_w, T^{diag}_w	Damped Picard iteration, isotropic / diagonal	Eq. (13)
ϵ_{rel}	Relative margin of the isotropic / diagonal cap	§6.2
$\mathcal{C}_s$, $\mathcal{C}^{\mathrm{diag}}_s$	Anti-collapse/positivity cap operator, isotropic / diagonal	Eq. (14)
r, r_j, r_0	Anti-collapse tolerance divisor / floor, coordinatewise / isotropic	Eq. (14)
$\mathcal{I}_s$, $\mathcal{I}_{s,j}$	Compact interval confining the iterates	Eq. (15)
L_Φ	Lipschitz constant of the Picard map Φ	Thm. 6.14
κ	Picard contraction modulus	Thm. 6.14
$\vartheta^\dagger$	Fixed point of the damped Picard iteration T_w	Thm. 6.14
m, M, m_ℓ, M_ℓ	Min / max curvature, global / per block	§6.3
ψ, ψ_ℓ	LRD regulariser, global / per block	Eq. (16)
$\lambda_{\min}(\cdot)$, $\lambda_{\max}(\cdot)$	Smallest / largest eigenvalue of a symmetric matrix	Lemma 6.19
a	Additive safeguard margin, $s = +1$	Thm. 6.20
φ	Safeguard scale factor, $s = -1$, $\varphi > 1$	Thm. 6.21
$u^2_{\max}$, $u^2_{\max,\ell}$	Feasibility threshold, global / per block	Rem. 6.22
$\xi \in (0,1)$	Relative feasibility margin for $u^2_{\max} = (1-\xi)/M$ ($s=-1$)	Rem. 6.22
ρ	Fixed SAM radius (distinct from learning rate ρ_t)	Eq. (18)
ε	SAM perturbation	Eq. (18)
$\alpha \in (0,1)$	Level-set threshold	Def. 6.4
ρ_α	Induced radius at level α	Eq. (21)
$u^\star_{\mathrm{soft}}$	Perturbation of the Gaussian candidate	Prop. 7.2
$\rho^\star(\eta)$	Learned (adaptive) SAM radius	Eq. (23)
K	Number of inner Picard steps	§7.2
s_{e}, $s_{\mathrm{e},\ell}$	Effective sign after the warm-up flip, global / per block	Alg. 3
b	Sherman–Morrison scalar in the $O(d)$ solve	Prop. 7.3

3 Background

This section fixes notation and recalls the constructions the two contributions rest on. Readers familiar with [1] may skip to Section 4.

3.1 Variational inference and Donsker–Varadhan

Let $n \in \mathbb{N}$, let $x_{1:n} = (x_1, \dots, x_n)$ be observations valued in $\mathcal{X} \subseteq \mathbb{R}^k$ for some $k \in \mathbb{N}$, $\{p(\cdot \mid \theta)\}_{\theta \in \Theta}$ a parametric model with $\Theta \subseteq \mathbb{R}^d$, and π a prior probability measure on Θ. Bayes' rule gives $\pi(\mathrm{d}\theta \mid x_{1:n}) \propto p(x_{1:n} \mid \theta)\, \pi(\mathrm{d}\theta)$, with the evidence $p(x_{1:n}) = \int_\Theta p(x_{1:n} \mid \theta)\, \pi(\mathrm{d}\theta)$ intractable outside conjugate models. Variational inference [10] replaces the posterior by the closest element of a tractable family $\mathcal{Q} \subset \mathcal{P}(\Theta)$ in the reverse Kullback–Leibler (KL) sense, where for $q \ll p$, $\mathrm{KL}(q\|p) \coloneqq \int \log(\mathrm{d}q/\mathrm{d}p)\, \mathrm{d}q$ (and $+\infty$ otherwise), via the decomposition

$$\log p(x_{1:n}) = \mathrm{ELBO}(q) + \mathrm{KL}\big(q \,\|\, \pi(\cdot \mid x_{1:n})\big), \qquad \mathrm{ELBO}(q) \coloneqq -\mathbb{E}_q[\ell(\theta)] - \mathrm{KL}(q\|\pi),$$

with $\ell(\theta) \coloneqq -\log p(x_{1:n} \mid \theta)$ the negative log-likelihood. Maximising the ELBO over $q \in \mathcal{Q}$ is therefore equivalent to minimising $\mathrm{KL}(q \,\|\, \pi(\cdot \mid x_{1:n}))$ without computing the evidence.

The structural result underlying both this and its possibilistic analogue is the Donsker–Varadhan variational formula [11]: for probability measures with $q \ll p$,

$$\mathrm{KL}(q\|p) = \sup_f \Big\{ \mathbb{E}_q[f] - \log \mathbb{E}_p\big[e^f\big] \Big\}, \tag{1}$$

the supremum ranging over measurable $f : \Theta \to \mathbb{R}$ with $\mathbb{E}_p[e^f] < \infty$, attained at $f^\star = \log(\mathrm{d}q/\mathrm{d}p)$. Equivalently, for a reference measure ν and measurable h with $\int e^h \, \mathrm{d}\nu < \infty$,

$$\log \int_\Theta e^{h(\theta)}\, \nu(\mathrm{d}\theta) = \sup_{\rho \in \mathcal{P}(\Theta)} \Big\{ \int h \, \mathrm{d}\rho - \mathrm{KL}(\rho\|\nu) \Big\}, \tag{2}$$

where $\mathcal{P}(\Theta)$ denotes the set of probability measures on Θ, attained at the Gibbs measure $\nu_h \propto e^h \nu$. Setting $h = -\ell$ and $\nu = \pi$ directly recovers the ELBO and identifies the Bayesian posterior as its unique maximiser. Setting $h = -\beta \hat{R}_n$, with $\hat{R}_n$ the empirical risk and $\beta > 0$ an inverse temperature, gives the technical core of the PAC-Bayes bound [12]. Equation (2) is the identity whose maxitive analogue drives everything below.

3.2 Possibility functions and possibilistic inference

Definition 3.1 (Possibility function, OPM [13, 14])**.** *A possibility function on Θ is a measurable $f : \Theta \to [0,1]$ with $\sup_{\theta \in \Theta} f(\theta) = 1$. We write $\mathcal{F}(\Theta)$ for the set of these. The associated outer probability measure (OPM) is $\bar{P}(A) \coloneqq \sup_{\theta \in A} f(\theta)$, and $\bar{P}(A)$ is the credibility of $\theta^\star \in A$, where $\theta^\star \in \Theta$ denotes the fixed, unknown quantity of interest introduced in Section 1.*

There is a formal correspondence with imprecise probability [15]: a possibility function characterises the closed convex set of probability measures it upper-bounds, $\{P : P(A) \le \sup_{\theta \in A} f(\theta),\ \forall A\}$, so possibilistic modelling is a rigorous instance of credal-set reasoning. Robust inference in complex models on this basis is developed in [7].

The set $\mathcal{F}(\Theta)$ is partially ordered by the specificity order $f \preceq g \iff f(\theta) \le g(\theta)$ for all $\theta \in \Theta$. A smaller function carries more information. Its least specific element is $\mathbf{1}_\Theta$ (the constant function equal to 1 on Θ), and $(f \vee g)(\theta) \coloneqq \max\{f(\theta), g(\theta)\}$ is the least upper bound. Unlike a probability density, a possibility function is not unique: any $g \succeq f$ also describes the same uncertain variable, with less information. When unique, the possibilistic expectation is the mode, $\mathbb{E}^*[\theta]_f \coloneqq \arg\max_{\theta \in \Theta} f(\theta)$, and the possibilistic precision is the negative log-Hessian at that mode, $\bar{I}_f \coloneqq -\nabla^2 \log f\big|_{\mathbb{E}^*[\theta]_f}$, a purely local quantity, unlike the probabilistic Fisher information, which averages globally over $p(\cdot \mid \theta)$.

The normal possibility function $\mathcal{N}^*(\theta;\mu,\Sigma)=\exp(-\frac{1}{2}(\theta-\mu)^\top\Sigma^{-1}(\theta-\mu))$ is a first example, with $\mathbb{E}^*[\theta]=\mu$ and $\bar{I}_f=\Sigma^{-1}$. It is not a density: it integrates to $(2\pi)^{d/2}|\Sigma|^{1/2}$, not to one, and admits the degenerate precision $\Sigma^{-1}=0$, which recovers $\mathbf{1}_\Theta$ exactly. Possibilistic Bayes' rule reads

$$g^\star_{\max}(\theta)=\frac{e^{-\ell(\theta)}\pi(\theta)}{\sup_{\theta'}e^{-\ell(\theta')}\pi(\theta')}=\frac{e^{-\hat{\ell}(\theta)}}{Z_{\max}},\qquad Z_{\max}\coloneqq\sup_{\theta\in\Theta}e^{-\hat{\ell}(\theta)},$$

where the denominator is an optimisation rather than an integral, and is always well defined for loss functions bounded from below, with no integrability condition on π.

3.3 The maxitive Donsker–Varadhan formula of Singh et al. [1]: a dual pair of objectives

Theorem 3.2 (Maxitive Donsker–Varadhan [1])**.** *Let $\pi:\Theta\to(0,1]$ be a possibility function and $h\coloneqq-\ell$. Then*

$$\log Z_{\max}\equiv\log\sup_{\theta\in\Theta}e^{h(\theta)}\pi(\theta)=\sup_{g\in\mathcal{F}(\Theta)}\inf_{\theta\in\Theta}\Big\{h(\theta)-\log\tfrac{g(\theta)}{\pi(\theta)}\Big\}=\inf_{g\in\mathcal{F}(\Theta)}\sup_{\theta\in\Theta}\Big\{h(\theta)-\log\tfrac{g(\theta)}{\pi(\theta)}\Big\},$$

with $\arg\max_g\inf_\theta\{\cdots\}=\{g\in\mathcal{F}(\Theta):g\preceq g^\star_{\max}\}$ *and* $\arg\min_g\sup_\theta\{\cdots\}=\{g\in\mathcal{F}(\Theta):g^\star_{\max}\preceq g\}$.

This motivates the two Consistency Bound Objectives

$$\underline{\mathrm{CBO}}(g)\coloneqq\inf_\theta\Big\{-\ell(\theta)-\log\tfrac{g(\theta)}{\pi(\theta)}\Big\},\qquad\overline{\mathrm{CBO}}(g)\coloneqq\sup_\theta\Big\{-\ell(\theta)-\log\tfrac{g(\theta)}{\pi(\theta)}\Big\},\tag{3}$$

satisfying $\underline{\mathrm{CBO}}(g)\le\log Z_{\max}\le\overline{\mathrm{CBO}}(g)$ for every $g\in\mathcal{F}(\Theta)$.

Two structural departures from probabilistic VI follow immediately. The variational characterisation is dual, with a lower and an upper objective where the ELBO has only a lower bound. The optimisers, moreover, are not unique. Here $\arg\max_g\underline{\mathrm{CBO}}(g)$ is the whole specificity interval $\{g\preceq g^\star_{\max}\}$, and $\arg\min_g\overline{\mathrm{CBO}}(g)$ is the whole interval $\{g^\star_{\max}\preceq g\}$. The posterior is recovered as the unique element of the intersection, $\{g^\star_{\max}\}=\arg\max_g\underline{\mathrm{CBO}}(g)\cap\arg\min_g\overline{\mathrm{CBO}}(g)$.

Remark 3.3 (Why the maxitive formula is dual). This duality is not an artefact of replacing $\int$ by sup in the probabilistic formula (2): it comes from the shape of Theorem 3.2 itself. The theorem states a min-max equality, $\sup_g\inf_\theta\{\cdots\}=\inf_g\sup_\theta\{\cdots\}=\log Z_{\max}$, and each side is read off as its own variational characterisation: maximising the inner infimum gives $\underline{\mathrm{CBO}}$, minimising the inner supremum gives $\overline{\mathrm{CBO}}$. Nothing forces these two readings to coincide pointwise in g away from the saddle, which is why two objectives are needed rather than one. The probabilistic formula (2) has no such second reading: $\mathrm{KL}(q\|p)$ is a single divergence of q from the fixed target p, and the Gibbs variational formula is already stated as one supremum over ρ, with no infimum-based formula computing the same $\log Z_{\mathrm{add}}$.

Zellner's theorem [16] shows that Bayes' rule is the unique information-processing rule that neither creates nor destroys information, so the probabilistic Donsker–Varadhan formula recovers the Bayesian posterior because it is optimal by this criterion, not by coincidence. Following an intuition suggested by Badr-Eddine Chérief-Abdellatif, [17] ask the same question for possibilistic updates, but reformulate the criterion itself: rather than Zellner's scalar measure of information, they conserve a pointwise information score, comparing candidates through the specificity order $\preceq$. $g^\star_{\max}$ is what that criterion identifies, and Theorem 3.2 expresses this same optimality result as a variational formula.

The two objectives are tied together by the max-relative entropy

$$D_{\max}(g\|f)\coloneqq\sup_{\theta\in\mathrm{supp}(f)}\log\frac{g(\theta)}{f(\theta)}\ge0,\qquad D_{\max}(g\|f)\coloneqq+\infty\ \text{ if }\ \mathrm{supp}(g)\not\subseteq\mathrm{supp}(f),\tag{4}$$

which vanishes exactly when $g\preceq f$, through the sandwich decomposition

$$\log Z_{\max}=\underline{\mathrm{CBO}}(g)+D_{\max}(g\|g^\star_{\max})=\overline{\mathrm{CBO}}(g)-D_{\max}(g^\star_{\max}\|g).\tag{5}$$

Maximising $\underline{\mathrm{CBO}}$ therefore minimises $D_{\max}(g\|g^\star_{\max})$, pushing g below the posterior (mode-seeking, under-estimation). Minimising $\overline{\mathrm{CBO}}$ minimises $D_{\max}(g^\star_{\max}\|g)$, pushing g above it (support-covering, over-estimation), as illustrated below in Figure 2 (Theorem 3.2, Eq. (5)).

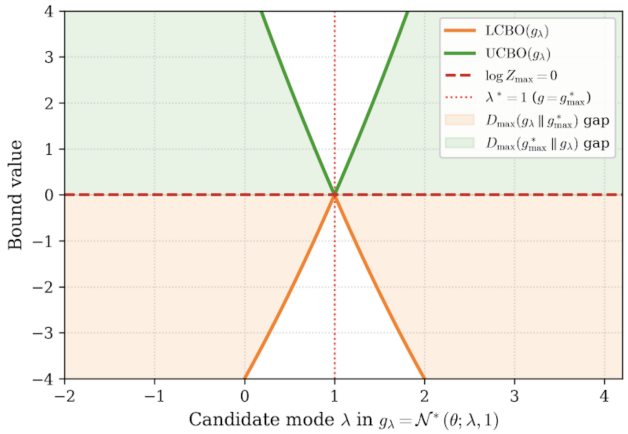


Figure 2: Sandwich decomposition (Theorem 3.2 and Equation (5)) for a Gaussian candidate $g_\lambda = \mathcal{N}^*(\cdot; \lambda, 1)$ against the target posterior $g^\star_{\max} = \mathcal{N}^*(\cdot; 1, 1)$ on $\Theta = [-3, 5]$. The lower bound $\underline{\mathrm{CBO}}(g_\lambda)$ and upper bound $\overline{\mathrm{CBO}}(g_\lambda)$ sandwich $\log Z_{\max} = 0$, intersecting tightly at the true mode $\lambda^\star = 1$.

Table 2 summarises the correspondence between the two frameworks.

Table 2: Structural correspondence between probabilistic and possibilistic variational inference.

	Probabilistic VI	**Possibilistic VI**
Candidate space	$\mathcal{P}(\Theta)$ $(\int q = 1)$	$\mathcal{F}(\Theta)$ $(\sup g = 1)$
Uncertainty	aleatoric	epistemic
Non-informative prior	improper $(\int = \infty)$	$\mathbf{1}_\Theta$, proper
Log-evidence	$\log Z_{\mathrm{add}} = \log \int e^{-\hat{\ell}(\theta)}\, \mathrm{d}\theta$	$\log Z_{\max} = \log \sup e^{-\hat{\ell}(\theta)}$
Variational formula	$\log Z_{\mathrm{add}} = \sup_q \mathrm{ELBO}(q)$	$\log Z_{\max} = \sup_g \underline{\mathrm{CBO}}(g) = \inf_g \overline{\mathrm{CBO}}(g)$
Divergence	$\mathrm{KL}(q \| \pi(\cdot \mid x_{1:n}))$	$D_{\max}(g \| g^\star_{\max})$ and $D_{\max}(g^\star_{\max} \| g)$
$D = 0 \iff$	$q = \pi(\cdot \mid x_{1:n})$	$g \preceq g^\star_{\max}$ and $g^\star_{\max} \preceq g$
Unique optimiser?	yes	no, intervals $\{g \preceq g^\star_{\max}\}$ and $\{g^\star_{\max} \preceq g\}$
Expectation	mean $\int \theta q(\mathrm{d}\theta)$	mode $\arg\max_\theta g(\theta)$
Fisher information	$\mathbb{E}[\nabla \log p \nabla \log p^\top]$ (global)	$-\nabla^2 \log g\|_{\mathrm{mode}}$ (local)

3.4 Possibilistic exponential families and the CBOpt updates of [1]

A possibilistic exponential family is $g_\lambda(\theta) = \exp(\lambda^\top T(\theta) - A(\lambda) - B(\theta))$ with $A(\lambda) = \sup_\theta[\lambda^\top T(\theta) - B(\theta)]$ a Legendre–Fenchel transform rather than a log-integral, convex, and finite as soon as $\lambda^\top T(\theta) - B(\theta)$ is bounded above in θ. This mirrors the Legendre duality of probabilistic exponential families [18], with A playing the role of the log-partition (the normalising constant, in log form).

The two CBOs are unified by a sign variable $s \in \{+1, -1\}$, the target being $\min_g s\, \mathrm{CBO}_s(g)$ in both cases,

$$\mathrm{CBO}_s(g) \coloneqq \operatorname*{ext}_\theta{}_s \Big\{ -\ell(\theta) - \log \tfrac{g(\theta)}{\pi(\theta)} \Big\}, \qquad \mathrm{ext}_{+1} = \sup, \quad \mathrm{ext}_{-1} = \inf .$$

Reparametrising by the mode $\eta = \theta^\star(\lambda) = \nabla A^\dagger(\lambda)$, where $A^\dagger(\lambda) \coloneqq \sup_\theta[\lambda^\top \theta - A(\theta)]$ is the Legendre transform of A, and defining the signed curvature $H_s(\eta) \coloneqq s(\nabla^2 \hat{\ell}(\eta) - \nabla^2 A_0(\eta))$, where A_0 denotes the log-partition of the specific candidate used in the update rather than the generic A above, and $\theta_s \coloneqq \operatorname*{argext}_\theta{}_s \{-\ell(\theta) - \log \frac{g(\theta)}{\pi(\theta)}\}$ the inner extremiser, Danskin's theorem gives $\nabla_\eta \mathrm{CBO}_s(\eta) = \nabla^2 A_0(\eta)(\eta - \theta_s)$. Replacing the inner objective by its second-order Taylor expansion around η and minimising the resulting surrogate of $s\, \mathrm{CBO}_s$ via natural gradient descent yields the signed Newton-like update, valid when $H_s \succ 0$,

$$\eta_{t+1} = \eta_t - \rho_t\, H_s(\eta_t)^{-1} \nabla \hat{\ell}(\eta_t). \tag{6}$$

With an isotropic Gaussian candidate of precision ϑ and an L^2 prior of precision δ (i.e. $-\log \pi(\theta) = \frac{\delta}{2}\|\theta\|^2$, the weight-decay term standard in neural-network training), the signed curvature becomes $H_s = s(\nabla^2 \ell + (\delta - \vartheta)I)$. Since the Hessian $\nabla^2 \ell$ is not available in closed form for neural-network objectives, we approximate its diagonal by an exponential moving average (EMA) of the squared stochastic gradient, $h_t \leftarrow \beta_2 h_{t-1} + (1-\beta_2)\hat{g}_t \odot \hat{g}_t$, where $\odot$ denotes the elementwise (Hadamard) product, with $\hat{g}_t$ a stochastic gradient at η_t and $\beta_2 \in [0, 1)$. This diagonal curvature proxy is the same one used by Adam, Vprop, and IVON. Using h_t in place of $\nabla^2 \ell(\eta_t)$ in H_s and in the update (6) gives the implementable form

$$\eta_{t+1} = \eta_t - \rho_t \, s \, \frac{\bar{g}_t + \delta \eta_t}{h_t + \delta - \vartheta}, \tag{7}$$

where $\bar{g}_t \coloneqq g_t/(1-\beta_1^t)$ is a bias-corrected EMA of the gradient itself ($g_t \leftarrow \beta_1 g_{t-1} + (1-\beta_1)\hat{g}_t$, $\beta_1 \in [0, 1)$, the correction removing the initialisation bias towards 0), and the division in the fraction is taken coordinatewise, since H_s is diagonal here.

Setting $s = +1$ and $\vartheta \in (0, \delta]$ gives `uCBOpt`. The adaptive variants set $\vartheta_t = \gamma c_t$ for an envelope c_t of the curvature estimate $\tilde{h}_t \coloneqq h_t + \delta$, chosen so that the sign requirement $H_s \succ 0$ holds pointwise (formalised later in Assumption 6.1). For $s = +1$ this requires $\vartheta_t < \tilde{h}_t$, ensured by taking $\gamma \in [0, 1)$ and $c_t = \min\{\beta_3 c_{t-1}, \tilde{h}_t\}$ ($\beta_3 > 1$), a growing minimum staying at or below $\tilde{h}_t$, giving `uCBOpt-adapt`. For $s = -1$ this requires $\vartheta_t > \tilde{h}_t$, ensured by taking $\gamma > 1$ and $c_t = \max\{\beta_3 c_{t-1}, \tilde{h}_t\}$ ($\beta_3 \in [0, 1)$), a decaying maximum staying at or above $\tilde{h}_t$, giving `lCBOpt-adapt`.

Remark 3.4 (Relation to Adam and IVON). Equation (7) matches Adam [19] except that Adam takes a square root of the second moment. Adam's root normalises by an estimate of the gradient's standard deviation, unrelated to any second-order model, whereas here the update is a Newton step $H_s^{-1}\nabla\hat{\ell}$, so h_t enters through a single matrix inverse and no root appears. With $\vartheta \to 0^+$ the candidate's curvature term drops from the denominator, leaving $h_t + \delta$ alone, the same form IVON [8] uses, though without its parameter perturbation step.

4 Reproducing [1]: `uCBOpt` at Fixed Precision

Extending the framework of [1] first required reproducing its core findings using the authors' provided implementation. This section reports a focused reproduction study, validating the metric and optimiser implementations on which Section 7 builds, and providing the baseline numbers reused directly in Table 10, with **repository link** in Appendix A.

Remark 4.1 (From the abstract framework to a neural network). $\theta \in \Theta \subseteq \mathbb{R}^d$ is the weight vector, d the parameter count (44,426 below). $\ell(\theta)$ is the cross-entropy loss, $\hat{\ell} = \ell - \log \pi$ adds the L^2 prior $\frac{\delta}{2}\|\theta\|^2$, and $g^\star_{\max}$ is never computed directly since that would require the global minimum of the loss. `CBOpt` uses mini-batch gradients of $\hat{\ell}$ to move $g_{\eta,\vartheta}$ towards $g^\star_{\max}$, η tracking the weights and ϑ the confidence around them.

4.1 Setup

Architecture. LeNet-5 variant: two convolutional layers ($1 \to 6$ then $6 \to 16$, 5×5 kernels), each followed by 2×2 max-pooling and ReLU activations, followed by three fully-connected layers ($256 \to 120 \to 84 \to 10$). This yields 44,426 trainable parameters, with dropout disabled.

Data. Fashion-MNIST with its 60,000 training images split 90/10 for training and validation (54,000 and 6,000 examples), alongside the standard 10,000 test images, using batch size 128 without data augmentation. EMNIST-Letters (14,800 test images) serves as the OOD evaluation set.

Training. 100 epochs with linear learning-rate warmup over the first five epochs followed by cosine decay to zero, no early stopping, evaluating the checkpoint with the lowest validation loss. Results are averaged over three seeds $(0, 1, 2)$.

Optimisers. All optimisers use $\beta_1 = 0.9$. We compare AdamW ($\rho = 10^{-3}$, $\delta = 10^{-2}$), IVON [8] ($\rho = 0.2$, $\delta = 2 \times 10^{-3}$, $\beta_2 = 0.99999$, $h_0 = 0.5$, effective sample size 5×10^4, one MC sample per training step), and `uCBOpt` ($\rho = 10^{-2}$, $\delta = 2 \times 10^{-3}$, $h_0 = 0.05$, $\vartheta_0 = 8 \times 10^{-6}$).

4.2 Metrics

All metrics use the checkpoint with the lowest validation loss. Accuracy, NLL, and ECE are in-domain, on Fashion-MNIST. FPR@95 and AUROC are OOD, Fashion-MNIST against EMNIST-Letters, using max-softmax confidence $\max_k \text{softmax}(\text{logits})_k$. AdamW and `uCBOpt` use a single deterministic pass. IVON uses its standard 64-sample Monte-Carlo ensemble [8], matching the protocol of [1], softmax applied per draw before averaging.

- **Accuracy.** Fraction of correctly classified test examples.
- **NLL.** With $p_k(x) \coloneqq \text{softmax}(\text{logits}(x))_k$ and y_i the true label,

$$\text{NLL} = -\frac{1}{n}\sum_{i=1}^{n} \log p_{y_i}(x_i).$$

  It penalises a confidently wrong prediction far more than an unsure one.

- **ECE.** Over 15 equal-width confidence bins $B_1, \ldots, B_{15}$,

$$\text{ECE} = \sum_{m=1}^{15} \frac{|B_m|}{n} \big|\text{acc}(B_m) - \text{conf}(B_m)\big|,$$

  with $\text{acc}(B_m)$ the empirical accuracy and $\text{conf}(B_m)$ the mean confidence in bin B_m. It measures discrepancies between stated confidence and actual accuracy.

- **FPR@95.** Following [1]'s discretisation, with $\text{TPR}(\tau)$, $\text{FPR}(\tau)$ the rates at threshold τ, FPR@95 is the mean $\text{FPR}(\tau)$ over `roc_curve` points with $\text{TPR}(\tau) \leq 0.9505$. It penalises unfamiliar inputs assigned high confidence.

- **AUROC.** Area under the ROC curve, integrating true against false positive rate across all τ. It quantifies the overall separation between in-distribution and OOD confidence.

4.3 Results

Table 3: Reproduction of [1] on LeNet-5/Fashion-MNIST with EMNIST as OOD set. Mean ± std over 3 seeds. Reference values below the line, best mean value per block in bold.

Optimiser	Acc. (%) ↑	NLL ↓	ECE ↓	FPR@95 ↓	AUROC ↑
`uCBOpt` [1]	90.9 ± 0.2	0.272 ± 0.004	0.025 ± 0.001	0.263 ± 0.005	0.795 ± 0.005
IVON [8]	$\mathbf{91.0 \pm 0.2}$	$\mathbf{0.258 \pm 0.003}$	$\mathbf{0.019 \pm 0.001}$	$\mathbf{0.241 \pm 0.009}$	$\mathbf{0.812 \pm 0.012}$
AdamW	89.6 ± 0.3	0.296 ± 0.005	0.022 ± 0.006	0.335 ± 0.005	0.722 ± 0.016
Reference values [1]					
uCBOpt	90.7 ± 0.2	0.270 ± 0.002	0.023 ± 0.000	$\mathbf{0.245 \pm 0.010}$	$\mathbf{0.810 \pm 0.012}$
IVON	$\mathbf{91.1 \pm 0.3}$	$\mathbf{0.258 \pm 0.001}$	$\mathbf{0.013 \pm 0.004}$	0.249 ± 0.024	0.804 ± 0.030
AdamW	90.0 ± 0.5	0.294 ± 0.007	0.026 ± 0.004	0.321 ± 0.008	0.745 ± 0.009

AdamW is the standard non-Bayesian baseline, decoupled weight decay and no candidate distribution [20]. IVON targets an additive Gaussian posterior, `uCBOpt` a normal possibility distribution, close enough that Remark 3.4 recovers IVON's update as the $\vartheta \to 0^+$ limit. IVON learns both the mode and a diagonal variance proxy, `uCBOpt` here fixes ϑ, learning only η, the limitation Section 6 addresses.

Every metric/optimiser pair matches the reference within two standard deviations, confirming each baseline reproduces faithfully. What differs is the ranking between IVON and `uCBOpt` on OOD detection: the reference has `uCBOpt` ahead on FPR@95 and AUROC by 0.004 and 0.006, below 0.2σ and not significant, while this reproduction places IVON marginally ahead. IVON's own reference standard deviations on these two metrics (0.024 and 0.030) exceed `uCBOpt`'s (0.010 and 0.012), so this reversal sits well within IVON's own noise, not a flaw in the implementation.

Remark 4.2 (Single-pass inference versus Monte-Carlo ensembling). IVON, a variational Bayesian optimiser targeting a Gaussian posterior, relies on Bayesian model averaging at test time, sampling 64 forward passes to integrate over parameter uncertainty. `uCBOpt` instead targets a normal possibility distribution whose prediction is given directly by the mode η, a single deterministic pass. Despite this 64× reduction in inference cost, `uCBOpt` matches IVON within two standard deviations across all five metrics, a practical argument for the possibilistic route.

5 Landscapes on Tractable Posteriors: A Diagnostic Bridge

This section investigates the behaviour of the lower and upper CBO objectives when the exact target $g^\star_{\max}$ is known analytically, serving as a diagnostic step prior to extending the candidate variational family in Section 6. §5.1 analyses four one-dimensional problems: three admitting closed-form expressions (a Gaussian sanity check, a Bernoulli–Beta model, and a Laplace distribution) and a symmetric bimodal mixture evaluated numerically. For each, we derive $g^\star_{\max}$, fit the Gaussian, Laplace, and Beta candidate families (defined below) where applicable, and evaluate the resulting bounds and divergences under the sandwich decomposition (5). §5.2 compiles the resulting gaps across all four cases. §5.3 then investigates whether these gaps come from the intrinsic geometry of $D_{\max}$ itself or from family misspecification, by relaxing $D_{\max}$ into an order-p divergence.

The three candidate families considered are the Gaussian $\mathcal{N}^*(\theta;\mu,\sigma^2) = \exp(-\frac{1}{2}\sigma^{-2}(\theta-\mu)^2)$ (the setting analysed in [1]), the Laplace $\mathrm{Lap}^*(\theta;\mu,b) = \exp(-|\theta-\mu|/b)$, and the Beta $\mathrm{Beta}^*(\theta;\alpha,\beta) \propto \exp((\alpha-1)\log\theta + (\beta-1)\log(1-\theta))$, normalised to unit supremum. The Gaussian and Beta candidates follow the possibilistic exponential-family form $g_\lambda(\theta) = \exp(\lambda^\top T(\theta) - A(\lambda) - B(\theta))$ introduced in Section 3. The Laplace candidate belongs to this class only with respect to b once μ is fixed, since $|\theta-\mu|$ depends directly on the location. Restricting the analysis to this class was originally intended to benchmark `CBOpt` runs against the exact reference solutions obtained here, an investigation left for future work to focus directly on the learned precisions of Section 6.

Each candidate family is fitted to $g^\star_{\max}$ over a discretised grid of $N = 500$ points, whose domain and grid spacing δ differ across cases (Table 4). For the Gaussian and Laplace candidates, the location μ is selected as the grid point maximising $g^\star_{\max}$. For the Gaussian scale, σ is obtained from the curvature of $\log g^\star_{\max}$ at that mode via a second-order central finite difference across the two adjacent grid points. For the Laplace scale, b is computed from the grid width where $g^\star_{\max} \geq 1/2$, divided by $2\log 2$. The Beta candidate has no separate location parameter and is fitted across all $N = 500$ points by ordinary least-squares regression on the log-linear model $\log g^\star_{\max}(\theta) \approx c + (\alpha-1)\log\theta + (\beta-1)\log(1-\theta)$, where the intercept c absorbs the normalisation constant. This model is exact since the log-possibility is strictly linear in $\log\theta$ and $\log(1-\theta)$.

Table 4: Grid domain and spacing for the four cases ($N = 500$ points throughout). Domain truncation affects the three cases on $\Theta = \mathbb{R}$, but not the Bernoulli–Beta case, whose domain $\Theta = (0,1)$ is already bounded.

Case	Domain rule	Θ range	δ
Gaussian (sanity)	$5\max(\sigma^\star/\sqrt{n}, \sigma_0)$ margin around $[\mu^\star, \mu_0]$	$[-10.0, 11.5]$	0.0431
Bernoulli–Beta	Fixed boundary cutoffs	$(10^{-6}, 1-10^{-6})$	0.0020
Laplace	$\pm 6b^\star$ around the sample median	$[-4.77, 7.23]$	0.0240
Bimodal mixture	Fixed bounds	$[-5.0, 5.0]$	0.0200

A fully unconstrained fit jointly optimising $\underline{\mathrm{CBO}}$ or $\overline{\mathrm{CBO}}$ over mode and scale was initially explored but abandoned: while the mode consistently settled near the discretised argmax regardless of scale, direct $\underline{\mathrm{CBO}}$ maximisation over scale collapses to a quasi-Dirac candidate ($\sigma \to 0$), which is an early sign of the precision degeneracy resolved in Section 6. Furthermore, a balanced $\underline{\mathrm{CBO}} - \overline{\mathrm{CBO}}$ criterion shifts the fitted mode away from the true mode instead of preventing collapse. The deterministic heuristics detailed above eliminate this collapse mode and are used throughout, keeping this section diagnostic.

Three effects of finite implementation choices recur below, worth separating once. The first is finite-sample noise: the observed statistic departs from the ground-truth value that generated the data, without consequence within the model, since the posteriors and fits above target the observed sample by construction, not the unknown ground truth. The second is grid resolution: a grid can fail to distinguish two nearby values close together in Θ. The third is domain truncation: whenever $\Theta = \mathbb{R}$ is unbounded, $\underline{\mathrm{CBO}}$ and $\overline{\mathrm{CBO}}$ are a true infimum and supremum over all of $\mathbb{R}$, not just the tested interval, and if a candidate does not match $g^\star_{\max}$ exactly, the inner objective can diverge as $|\theta| \to \infty$, so the grid value is bounded by where the grid stops rather than converged; $D_{\max}(g\|g^\star) = \log Z_{\max} - \underline{\mathrm{CBO}}(g)$ and $D_{\max}(g^\star\|g) = \overline{\mathrm{CBO}}(g) - \log Z_{\max}$ inherit this directly. Grid resolution and domain truncation are unavoidable, since some finite δ and domain must be chosen to evaluate the two objectives.

The comparisons below stay interpretable because they are relative and qualitative, by order of magnitude, and within each case all candidate families are fitted and evaluated on the same grid. Wherever a gap is small enough for either effect to plausibly explain it, it is checked directly, by refining δ or widening the domain to see whether the value moves. Corresponding **repository link** is available in Appendix A.

5.1 Four tractable posteriors: Gaussian, Bernoulli–Beta, Laplace, and a bimodal mixture

Case 1: Gaussian likelihood, known variance (sanity check). $\Theta = \mathbb{R}$, with the single parameter to infer being the mean $\theta = \mu$. The observation noise scale $\sigma^\star = 1$ is fixed as a known constant of the data-generating model, and the prior possibility function is Gaussian, $\pi = \mathcal{N}^*(\cdot; \mu_0, \sigma_0^2)$ with $\mu_0 = 0, \sigma_0 = 2$. We simulate $n = 30$ realisations $x_{1:n} \sim \mathcal{N}(\mu^\star, \sigma^{\star 2})$, with ground-truth mean $\mu^\star = 1.5$, yielding a sample mean $\bar{x} = 1.2455$ (the first effect above, sampling noise alone, without consequence within the model). The conjugate posterior is available in closed form:

$$\sigma_{\mathrm{post}} = \Big(\frac{n}{\sigma^{\star 2}} + \frac{1}{\sigma_0^2}\Big)^{-1/2} = 0.1818, \qquad \mu_{\mathrm{post}} = \sigma^2_{\mathrm{post}}\Big(\frac{n\bar{x}}{\sigma^{\star 2}} + \frac{\mu_0}{\sigma_0^2}\Big) = 1.2352, \qquad \log Z_{\max} = -0.1961.$$

For a Gaussian candidate the inner objective is quadratic, so the CBO landscape is smooth and admits a single crossing point on this grid where $\underline{\mathrm{CBO}} = \overline{\mathrm{CBO}} = \log Z_{\max}$. The second effect, resolution, appears here: the grid-discretised mode $\hat{\mu} = 1.2455$ lands on $\bar{x}$ rather than the true $\mu_{\mathrm{post}} = 1.2352$. The prior is comparatively weak ($1/\sigma_0^2 = 0.25$ against $n/\sigma^{\star 2} = 30$ from the likelihood), so it shrinks μ_{post} only 0.0103 from $\bar{x}$, below the grid spacing ($\delta \approx 0.0431$), and the discretised mode search cannot resolve it. The curvature estimate $\hat{\sigma} = 0.1818$, by contrast, recovers σ_{post} to the four decimals shown regardless, since $\log g^\star_{\max}$ is a parabola everywhere on Θ and its curvature is the same constant wherever it is measured.

The reported $D_{\max}(g\|g^\star) \approx 0.57$ and $D_{\max}(g^\star\|g) \approx 0.50$ are not fully explained by this mode error: the third effect, domain truncation, is also at work, and neither $\underline{\mathrm{CBO}}$ nor $\overline{\mathrm{CBO}}$ has actually converged. The mode error above and a far smaller mismatch in σ (5×10^{-7}) both mean the fitted candidate is not exactly $g^\star_{\max}$, and on $\Theta = \mathbb{R}$ that suffices, since the two log-possibilities differ by a nonzero quadratic in θ, which grows without bound as $|\theta| \to \infty$ however small the mismatch. The θ that minimises (resp. maximises) the inner objective sits exactly at the grid boundary, $\theta = -10.0$ (resp. $\theta = +11.5$), and widening the domain at matched density keeps moving both values. Doubling it to $[-20, 21.5]$ gives $\underline{\mathrm{CBO}} = -1.2925$, $\overline{\mathrm{CBO}} = 0.7632$, and doubling again to $[-40, 41.5]$ gives $\underline{\mathrm{CBO}} = -2.3867$, $\overline{\mathrm{CBO}} = 1.6504$. Even in this best-specified case, resolution and truncation both leave their mark on the reported values.

This case's value as a sanity check is to confirm all three effects at once, with the candidate family correctly specified. Sampling noise separates $\bar{x}$ from $\mu^\star$ without consequence, resolution explains the mode error, and both that mode error and the far smaller residual mismatch in σ are each enough on an unbounded Θ to let truncation determine the values of the reported gap. It sets the reference against which the gaps of the other three cases are read.

Case 2: Bernoulli–Beta. $\Theta = (0,1)$, with the single parameter to infer being the success probability $\theta = p$. The prior possibility function is Beta, $\pi = \mathrm{Beta}^*(\alpha_0, \beta_0)$ with $\alpha_0 = \beta_0 = 2$. We simulate $n = 20$ realisations $x_{1:n} \sim \mathrm{Bernoulli}(p^\star)$, with ground-truth $p^\star = 0.7$, yielding a realised count of $s = 11$ successes, below the expectation $np^\star = 14$, the first effect, sampling noise, without consequence within the model. The conjugate posterior is available in closed form:

$$\alpha^\star = \alpha_0 + s = 13, \qquad \beta^\star = \beta_0 + (n-s) = 11, \qquad \theta^\star = \frac{\alpha^\star - 1}{\alpha^\star + \beta^\star - 2} = 0.5455, \qquad \log Z_{\max} = -15.1582.$$

The least-squares fit recovers $(\alpha^\star, \beta^\star) = (13, 11)$ exactly, with $D_{\max}(g\|g^\star) = 3.05 \times 10^{-5}$, $D_{\max}(g^\star\|g) = 2.29 \times 10^{-5}$ and $\underline{\mathrm{CBO}} = \overline{\mathrm{CBO}} = \log Z_{\max}$, so the sandwich is tight. The second effect, resolution, enters differently here than for a discretised mode: with no single point to locate, δ instead sets the density of points feeding the regression, and at $N = 500$ this density is already enough to recover the exact shape parameters. The third effect, truncation, cannot apply at all, since $\Theta = (0,1)$ is bounded, so $\underline{\mathrm{CBO}}$ and $\overline{\mathrm{CBO}}$ are a true infimum and supremum over the whole domain already, not merely the grid tested. The residual order of magnitude ($\sim 10^{-5}$) reflects regression precision alone.

The Gaussian candidate, by contrast, gives $D_{\max}(g\|g^\star) = 137.33$ even after optimal fitting, and this is not a fitting failure but a support mismatch. The Gaussian assigns positive mass outside $(0,1)$, where $g^\star_{\max}$ vanishes, so $\log g_{\mathcal{N}^*} - \log g^\star_{\max}$ is unbounded on the boundary and the supremum diverges. This case shows two things at once. On the Beta side, none of the three finite-implementation effects has any impact, and the fit behaves exactly as expected. On the Gaussian side, the divergence matches exactly what the support mismatch predicted by theory.

Case 3: Laplace likelihood. $\Theta = \mathbb{R}$, with the single parameter to infer being the location $\theta = \mu$. The observation scale $b^\star = 1.0$ is fixed as a known constant, and the prior possibility function is uniform, $\pi = \mathbf{1}_\Theta$, so the regularised loss is $\hat{\ell}(\theta) = \ell(\theta) = \sum_i |x_i - \theta|$. We simulate $n = 20$ realisations $x_{1:n} \sim \mathrm{Laplace}(\mu^\star, b^\star)$, with ground-truth location $\mu^\star = 1.0$, yielding a discretised sample median $\hat{\theta}_{\mathrm{med}} = 1.2374$, below $\mu^\star$ by sampling noise, the first effect, and accurate only to the grid spacing δ, the second effect. The posterior $g^\star_{\max}$ is piecewise linear in log-space, maximised at $\hat{\theta}_{\mathrm{med}}$, with $\log Z_{\max} = -17.0948$.

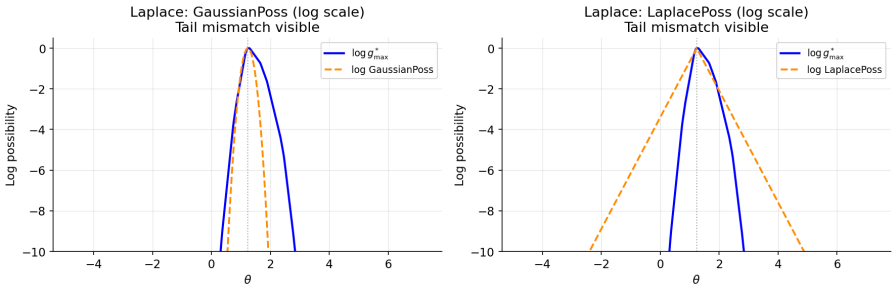


Figure 3: Case 3, log-scale view. Left, Gaussian candidate; right, Laplace candidate. $\log g^\star_{\max}$ decays linearly in the tails, faster than any single candidate can match, since it sums $n = 20$ Laplace terms. The Gaussian log-possibility decays quadratically, so $g^\star_{\max}/\mathcal{N}^* \to +\infty$ there. The fitted Laplace candidate decays too slowly to match either tail, leaving $D_{\max}(g_{\mathrm{Lap}^*}\|g^\star) = +\infty$ on $\mathbb{R}$ as well.

The Gaussian candidate (Figure 3, left) fails for a different reason than in Case 2. Far from the data, $\log g^\star_{\max}$ decays linearly while $\log \mathcal{N}^*$ remains a downward parabola, so $g^\star_{\max}/\mathcal{N}^* \to +\infty$ in the tails, forcing $D_{\max}(g^\star\|g_{\mathcal{N}^*}) = +\infty$ in the continuous limit (650.08 on the grid). This is a tail mismatch, not a support mismatch as in Case 2, and the third effect, domain truncation, is what keeps the reported value finite: no member of the Gaussian family escapes the divergence itself.

The Laplace candidate (Figure 3, right) reduces the tail mismatch relative to the Gaussian, but $D_{\max}(g_{\mathrm{Lap}^*}\|g^\star) = 93.89$ on this grid is itself a truncated value, the same third effect at work again. Widening the domain drives it further up without bound, since $\log g^\star_{\max}(\theta) \sim -n_{\mathrm{obs}}|\theta|$ decays faster than the fitted candidate's $-|\theta|/b_{\mathrm{fit}}$, so $D_{\max}(g_{\mathrm{Lap}^*}\|g^\star) = +\infty$ in the continuous limit, just as for the Gaussian above. This is not a failure of the Laplace family as such: summing $n = 20$ Laplace log-possibilities decays faster than any single Laplace candidate can match.

Case 4: bimodal posterior. $\Theta = \mathbb{R}$, with the single location parameter θ to infer, under the uniform prior $\pi = \mathbf{1}_\Theta$. We simulate $n = 10$ realisations $x_{1:n} \sim \frac{1}{2}\mathcal{N}(\theta^\star, 1) + \frac{1}{2}\mathcal{N}(-\theta^\star, 1)$, with true value $\theta^\star = 2$. This case is deliberately set up to fail, since the posterior needs two location parameters, one per mode, while every candidate here carries only one, μ. Each observation's likelihood is a two-Gaussian mixture, not exponential, so $g^\star_{\max}$ has no closed form: $\ell(\theta)$ is evaluated pointwise via log-sum-exp, giving $\log Z_{\max} = -9.4045$.

A first experiment fits the mode and curvature automatically, targeting the argmax of $g^\star_{\max}$ directly rather than $\underline{\mathrm{CBO}}$. It lands on the left mode, $\hat{\mu} = -2.13, \hat{\sigma} = 0.32$. Neither $\underline{\mathrm{CBO}} = -9.92$ nor $\overline{\mathrm{CBO}} = 200.82$ has converged: both worst points sit at the domain boundary and keep growing as the domain doubles, $\underline{\mathrm{CBO}}$ to -13.18 then -29.16 and $\overline{\mathrm{CBO}}$ to 408.29 then 813.78. The qualitative picture survives: this mode-seeking candidate stays close to $g^\star_{\max}$ near its own mode, so the small $D_{\max}(g\|g^\star)$ side is credible, while its failure to be mass-covering leaves $D_{\max}(g^\star\|g)$ large and growing rather than converging as the domain widens.

A second experiment fixes μ by hand at $\mathcal{N}^*(\cdot; 2, 0.8)$, on the mode the fit does not select, giving $\underline{\mathrm{CBO}} = -43.4301$, $\overline{\mathrm{CBO}} = 6.3974$. $\underline{\mathrm{CBO}}$ has not converged, growing to -268.62 then -1351.82 as the domain doubles, so $D_{\max}(g\|g^\star) = 34.03$ is itself grid-bound. $\overline{\mathrm{CBO}}$, by contrast, is stable across all three widths $(6.3974, 6.3972, 6.3972)$, so $D_{\max}(g^\star\|g) = 15.80$ is a converged gap: this candidate's wider $\sigma = 0.8$ already covers enough of its own mode and the valley for the over-coverage to peak at an interior point. This reverses which side diverges compared to the first experiment: a wider candidate spreads too much possibility into the far tails, where $g^\star_{\max}$ is zero, so mode-seeking is what breaks here instead.

Both experiments confirm a single Gaussian candidate cannot be mode-seeking and mass-covering at once here. Possibility functions admit an operation with no probabilistic counterpart, namely $(g_A \vee g_B)(\theta) \coloneqq \max(g_A(\theta), g_B(\theta))$, again a valid possibility function. Learning each component's mode and precision, as Section 6 intends to do, could fit each mode's local geometry closely, though whether that suffices to shrink both divergences depends on the match away from the modes, since $g^\star_{\max}$ is not itself a two-Gaussian mixture. Extending the construction of Section 6 to such a mixture, as discussed with Jérémie Houssineau, or using a non-parametric possibility candidate instead, are both left open.

5.2 Cross-case summary

Table 5: Sandwich decomposition (5) at fitted candidate parameters. $D_{\max}(g\|g^\star)$ is the under-estimation gap (LCBO residual), $D_{\max}(g^\star\|g)$ the over-estimation gap (UCBO residual).

Case	Family	$\log Z_{\max}$	$\underline{\mathrm{CBO}}$	$\overline{\mathrm{CBO}}$	$D_{\max}(g\|g^\star)$	$D_{\max}(g^\star\|g)$
Gaussian (sanity)	$\mathcal{N}^*$	-0.1961	-0.7632	0.3020	0.5671	0.4980
Bernoulli–Beta	$\mathcal{N}^*$	-15.1582	-152.4862	-15.1556	137.3280	0.0026
	Beta*	-15.1582	-15.1582	-15.1582	3.05×10^{-5}	2.29×10^{-5}
Laplace	$\mathcal{N}^*$	-17.0948	-17.4412	632.9875	0.3464	650.0823
	Lap*	-17.0948	-110.9888	-16.6765	93.8941	0.4183
Bimodal	$\mathcal{N}^*$ (μ=2)	-9.4045	-43.4301	6.3974	34.0256	15.8019

Table 5 shows a clear pattern: wherever the family matches the structural shape of $g^\star_{\max}$, both $D_{\max}$ terms collapse to numerical noise, and wherever it does not, one side explodes to two to three orders of magnitude above the scale of $\log Z_{\max}$ itself. Only Bernoulli–Beta is immune to domain truncation, since $\Theta = (0, 1)$ is bounded, and gives the sharpest illustration of the pattern within a single case: the same target gives $D_{\max}(g\|g^\star) = 137.33$ for the Gaussian and 3.05×10^{-5} for the Beta candidate, a five-order-of-magnitude gap driven purely by the choice of family.

5.3 Is $D_{\max}$ the right divergence?

Domain truncation is what left several of the values above unconverged, since the max-relative entropy is a worst-case ratio in which a single problematic point can dominate the whole value. This invites the question whether a softer divergence would behave better, for example a p-th order variant recovering $D_{\max}$ as $p \to \infty$, a possibilistic analogue of Rényi VI [21] or of the χ upper bound [22].

Softening the divergence: an order-p experiment. We test the p-th order family, a direct one-parameter softening of $D_{\max}$ itself (log-sum-exp in place of the hard max), rather than the Rényi/χ analogue, which would swap in a differently-shaped functional and leave it unclear whether any change reflects the softening or the new functional. Define

$$D_p(g\|f) := \frac{1}{p}\log \sum_{\theta\in\mathrm{supp}(f)} \exp\big(p(\log g(\theta) - \log f(\theta))\big), \qquad D_p(g\|f) := +\infty \ \text{ if } \ \mathrm{supp}(g) \not\subseteq \mathrm{supp}(f), \tag{8}$$

which recovers $D_{\max}$ exactly as $p \to \infty$ and softens the hard supremum for finite p. Reusing the already-fitted Gaussian candidates g above, without refitting, Table 6 reports $D_p(g\|g^\star)$ and $D_p(g^\star\|g)$ at $p = 1$ (softest) and $p = 50$ against the $D_{\max}$ reference.

Table 6: Order-p divergence at $p = 1$ (softest) and $p = 50$, against the $D_{\max}$ reference of Table 5.

	Gaussian (sanity)		Bernoulli–Beta		Laplace		Bimodal	
	$D(g\|g^\star)$	$D(g^\star\|g)$	$D(g\|g^\star)$	$D(g^\star\|g)$	$D(g\|g^\star)$	$D(g^\star\|g)$	$D(g\|g^\star)$	$D(g^\star\|g)$
D_1	6.29	6.23	137.33	5.64	3.44	650.09	35.00	19.57
D_{50}	0.61	0.54	137.33	0.10	0.36	650.08	34.03	15.84
$D_{\max}$ (ref)	0.57	0.50	137.33	0.003	0.35	650.08	34.03	15.80

In the two clearest structural mismatches, support (Bernoulli–Beta) and tail (Laplace), the mismatched direction stays fixed (137.3280 → 137.3280 and 650.08 → 650.09) from $p = 50$ down to $p = 1$, matching $D_{\max}$ to four decimals; softening does not inflate these gaps at all. The well-specified direction in each case rises substantially as p drops instead, as does both directions of the Gaussian sanity check (0.61 → 6.29), a grid artefact rather than a structural one. The bimodal case is intermediate: the mode-seeking direction barely shifts (34.0 → 35.0), the support-covering direction rises more (15.8 → 19.6) but stays large.

By the standard log-sum-exp bound, $D_{\max} \le D_p \le D_{\max} + \frac{1}{p}\log N$ on an N-point grid, so D_p never falls below $D_{\max}$ and decreases towards it as p grows, tightly if one point dominates the sum and loosely if many contribute comparably. This is why a small D_1 is reliable evidence of a good fit but a large one is not, by itself, evidence of a mismatch: for Bernoulli–Beta and Laplace, one point already dominates at $p = 1$, so D_1 sits close to $D_{\max}$ regardless of the mismatch, whereas for the well-specified Gaussian sanity check, many comparable residuals inflate D_1 well above the true $D_{\max}$.

Reading a mismatch off a large D_1 therefore requires comparing D_1, D_{50}, and $D_{\max}$ to rule out an inflated well-specified value. That is not a problem here: the two orders of magnitude never overlap, even the most inflated well-specified value ($D_1 = 6.29$) stays far below the smallest mismatched one (137.33), so the ordering is the same at $p = 1$, $p = 50$, and $D_{\max}$ throughout Table 6. $D_{\max}$ is therefore not the wrong divergence for this diagnosis: softening changes the magnitude but never the verdict, so effort is better spent finding a better candidate family than tuning the divergence used to score it, the motivation for Section 6.

Remark 5.1 (Varying p during optimisation). Jérémie Houssineau mentioned in a discussion the possibility of varying p over the course of optimisation rather than fixing it, starting soft to keep the objective well behaved while the candidate is far from $g^\star_{\max}$ and hardening towards $D_{\max}$ as it converges. This is left open.

6 Constructing a Learned Precision

Section 5 showed that directly optimising the bound leads to a degeneracy: the objective collapses onto a boundary optimum, yielding a quasi-Dirac candidate ($\sigma \to 0$, $\vartheta \to \infty$) in the case of CBO. While the `CBOpt` family of Singh et al. [1] sidesteps this failure mode by keeping the candidate precision constant, this prevents the dispersion from adapting to the local geometry around the current mode η. This section resolves both issues simultaneously by learning the precision conditionally on η at each iteration (see Remark 6.17 regarding joint convergence).

Throughout, we denote the extremiser by $\theta_s^X(\cdot)$ and the Newton step by $u_s^X(\cdot) \coloneqq -H_s^X(\cdot)^{-1}\nabla\hat{\ell}(\eta)$ (with $\theta_s^X - \eta \approx s u_s^X$), across four precision structures under consideration: the isotropic case ($X = \emptyset$, $\vartheta \in \mathbb{R}_{>0}$), the diagonal case ($X = \mathrm{diag}$, $\boldsymbol{\vartheta} \in \mathbb{R}^d_{>0}$), the low-rank-plus-diagonal case ($X = \mathrm{LRD}$, $S^{-1} = u_s^{\mathrm{LRD}}(u_s^{\mathrm{LRD}})^\top + \psi I$, §6.3), and the block-diagonal case ($X = \mathrm{BD}$, $S = \bigoplus_\ell S_\ell$). We begin by investigating why a specificity penalty is necessary in the first place and what explicit form it should take (§6.1), before deriving closed-form fixed-point solutions together with the curvature safeguards ensuring $H_s \succ 0$ (§6.2, §6.3).

Standing assumption. The hypotheses below are what §6.1 relies on for the isotropic case, under an L^2 prior $-\log\pi(\theta) = \frac{\delta}{2}\|\theta\|^2$, so that $\nabla^2\hat{\ell} = \nabla^2\ell + \delta I$. These are subsequently extended by §6.2 and §6.3 to, respectively, the diagonal and the general matrix cases. §7.1 states its own hypotheses wherever they are needed, with the exception of its learned-radius paragraph, which inherits the present ones unchanged by directly reusing Prop. 6.10.

Assumption 6.1 (Local regularity, curvature, and inner-extremiser regularity)**.** *(a) ℓ is twice continuously differentiable on an open neighbourhood U of the current iterate η.*

(b) The signed Hessian satisfies $H_s(\theta,\vartheta) = s(\nabla^2\hat{\ell}(\theta) - \vartheta I) \succ 0$ for all $\theta \in U$ and all ϑ in an admissible interval $\mathcal{I}_s \subset (0,\infty)$.

(c) At fixed $\vartheta \in \mathcal{I}_s$, the inner extremiser $\theta_s(\eta)$ exists, is unique, lies in U, and depends continuously on η.

(d) At fixed η, for every $\vartheta \in \mathcal{I}_s$, the inner extremiser $\theta_s(\vartheta)$ exists, is unique, lies in U with $\theta_s(\vartheta) \neq \eta$ (so $u_s(\vartheta) = \theta_s(\vartheta) - \eta \neq 0$), and depends continuously on ϑ on $\mathcal{I}_s$.

Hypotheses (a)–(c) are already implicit in the work of Singh et al. [1]. Indeed, (a) is presupposed by their Taylor expansion of ℓ around η; (b), in turn, ensures that the inner objective is strictly concave for $s = +1$ (resp. strictly convex for $s = -1$) and guarantees that H_s is invertible in the update (6); and (c) is implicit in their application of Danskin's theorem with respect to η. Hypothesis (d), by contrast, is our main addition: it extends these properties to $\theta_s(\vartheta)$ at fixed η over an entire admissible interval $\mathcal{I}_s$, rather than merely at a single point, a situation absent from Singh et al. since ϑ never varies for them. It serves two purposes: it keeps $u_s(\vartheta) \neq 0$, needed for the division by u_s in Proposition 6.10 and in the damped Picard map of (13), and it makes Danskin's theorem applicable along the whole of $\mathcal{I}_s$, from which the first-order condition of Proposition 6.7 follows, its continuity in turn supplying the existence argument.

Each of these hypotheses is, moreover, reasonable in the present setting. The smoothness of ℓ required by (a) holds for common architectures, in particular almost everywhere for ReLU networks [23]. The condition $H_s \succ 0$ required by (b), on the other hand, is not automatic during training, which is why §6.2 introduces spectral caps for the isotropic and diagonal cases, later extended in §6.3 to the LRD and BD geometries. Confining $\theta_s(\eta)$ to U as required by (c) is plausible near a stationary point, where $u_s \approx -H_s(\eta)^{-1}\nabla\hat{\ell}(\eta) \to 0$; this plausibility does not, however, transfer to (d), where η is instead held fixed and $\|u_s(\vartheta)\|$ diverges as ϑ approaches the boundary of $H_s \succ 0$ (Lemma 6.6). Accordingly, §6.1 relaxes the requirement $\theta_s \in U$ in order to study the full variation of ϑ over $\mathbb{R}^d$, while the practical implementation instead relies on the safeguard caps of §6.2 to keep ϑ away from the spectral boundary and thereby retain $\theta_s(\vartheta) \in U$.

6.1 From degeneracy to a log-volume penalty

The raw objective is degenerate. Starting from the signed objective $\mathrm{CBO}_s(g)$ of Section 3.4, taking the isotropic Gaussian candidate $g_{\eta,\vartheta} = \mathcal{N}^*(\theta;\eta,\vartheta^{-1}I)$, the L^2 prior $-\log\pi(\theta) = \frac{\delta}{2}\|\theta\|^2$ as in [1] and Section 5 (kept here for consistency only, see the remark below), and letting ϑ vary as a free parameter of the objective itself gives

$$\mathrm{CBO}_s(\eta,\vartheta) \coloneqq \operatorname*{ext}_{\theta,s}\Big\{-\hat{\ell}(\theta) + \tfrac{\vartheta}{2}\|\theta-\eta\|^2\Big\}.$$

Remark 6.2 (The prior plays no role here). Whether π is this L^2 parameter prior or the uninformative $\pi \equiv 1$ makes no difference here, since $\hat{\ell}$ has no explicit dependence on ϑ and the prior drops out of every derivative with respect to ϑ taken below.

Section 5 already ran into the consequence of this, with maximising CBO_s directly collapsing the candidate to a quasi-Dirac spike ($\sigma \to 0$), an early sign of a degeneracy this section now makes precise.

Lemma 6.3 (Degeneracy of the raw objective)**.** *Under Assumption 6.1(d), $\vartheta \mapsto \mathrm{CBO}_s(\eta,\vartheta)$ is strictly increasing on its admissible domain for both signs, so the optimum sits only at the boundary ($\vartheta \to 0^+$ for the upper CBO minimisation and $\vartheta \to \infty$ for the lower CBO maximisation), and CBO_s cannot identify an interior value of the precision.*

Proof. Under Assumption 6.1(d), for any admissible ϑ, $\theta_s(\vartheta)$ is unique, interior to U, and satisfies $\theta_s(\vartheta) \neq \eta$. By Danskin's envelope theorem applied to the inner supremum ($s=+1$) and infimum ($s=-1$), $\vartheta \mapsto \mathrm{CBO}_s(\eta,\vartheta)$ is differentiable, with

$$\frac{\partial}{\partial\vartheta}\mathrm{CBO}_s(\eta,\vartheta) = \frac{1}{2}\|\theta_s(\vartheta)-\eta\|^2 = \frac{1}{2}\|u_s(\vartheta)\|^2 > 0.$$

Strictly positive everywhere, this makes $\mathrm{CBO}_s(\eta,\cdot)$ strictly increasing for both $s=+1$ and $s=-1$: minimising $\overline{\mathrm{CBO}}$ ($s=+1$) pushes $\vartheta \to 0^+$, while maximising $\underline{\mathrm{CBO}}$ ($s=-1$) pushes $\vartheta \to \infty$, precluding any interior stationary point. □

This is the precise sense in which the precision is unidentifiable without a penalty: the raw objective always collapses to a boundary, exactly what Section 5 observed when maximising $\underline{\mathrm{CBO}}$ over scale alone collapsed to a quasi-Dirac candidate ($\sigma \to 0$). Probabilistic VI never faces this gap, since the ELBO already carries an entropy term penalising concentration. One must be added here, since a boundary optimum gives no usable precision and no equation to build an update rule from. The log-volume penalty introduced below breaks this strict monotonicity for either sign.

A geometric penalty. For this paragraph and the specialisation to the four cases that closes this subsection, we work with a general precision matrix $S \in \mathcal{S}^d_{++}$ (the cone of symmetric positive-definite $d\times d$ real matrices) and $g_{\eta,S} = \mathcal{N}^*(\theta;\eta,S^{-1})$, of which the isotropic, diagonal, LRD, and BD candidates studied here are each a particular choice. We return to the isotropic L^2-prior form to show the penalised objective is no longer strictly monotone.

A penalty shaped like the probabilistic Gaussian log-normaliser $\frac{1}{2}\log\det(2\pi\Sigma)$ is not directly justified here, since a possibility function has no normaliser and nothing in the definition of g penalises concentration. Shelvia Wongso suggested a geometric answer instead: the missing cost is the log-volume of the candidate's own level sets. The results below make this precise and recover a term of the same shape.

Definition 6.4 (Level-set volume)**.** *For $g_{\eta,S} = \mathcal{N}^*(\theta;\eta,S^{-1})$ with $S \in \mathcal{S}^d_{++}$ and $\alpha \in (0,1)$, the level set $\{\theta : g_{\eta,S}(\theta) \geq \alpha\}$ is the ellipsoid $\{(\theta-\eta)^\top S(\theta-\eta) \leq -2\log\alpha\}$, of Lebesgue volume $\mathrm{vol}_\alpha(S) = V_d(-2\log\alpha)^{d/2}(\det S)^{-1/2}$, where $V_d = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$ denotes the volume of the Euclidean unit ball in $\mathbb{R}^d$. Hence*

$$\log\mathrm{vol}_\alpha(S) = -\tfrac{1}{2}\log\det S + c(\alpha,d), \tag{9}$$

with $c(\alpha,d)$ independent of S: the level α shifts the log-volume by an additive constant and never changes which candidate is smaller.

The log-volume penalty $-\frac{1}{2}\log\det S$ is therefore not an import from probability, but coincides exactly with $\log \mathrm{vol}_\alpha(S)$ up to the S-independent constant $c(\alpha, d)$, precisely what Lemma 6.3 showed the raw objective leaves unpenalised: a small volume means a concentrated, informative candidate, a large one an uninformative one. Recall that $\overline{\mathrm{CBO}}(g) \le \log Z_{\max}$ is equivalent to $g \ge g^\star$ pointwise, and $\underline{\mathrm{CBO}}(g) \ge \log Z_{\max}$ to $g \le g^\star$ pointwise. Any $g \ge g^\star$ is trivially satisfied by $g \equiv \mathbf{1}_\Theta$, so the useful such candidate is the smallest in volume, closest to $g^\star$ from above. Conversely, any $g \le g^\star$ is trivially satisfied by an arbitrarily narrow spike, so the useful one is the largest in volume, closest to $g^\star$ from below. Two questions remain: at what multiplier μ, if any, does the log-volume penalty coincide with the Lagrangian of this extremal-volume selection, and is it the same for both directions?

Writing out the candidate penalised objective in full,

$$\mathrm{PCBO}_s(\eta, S) = \operatorname*{ext}_\theta{}_s\Big\{-\hat{\ell}(\theta) + \tfrac{1}{2}(\theta-\eta)^\top S(\theta-\eta) - \tfrac{1}{2}\log\det S\Big\}, \tag{10}$$

we show that $-\frac{1}{2}\log\det S$ is precisely what this extremal-volume selection demands at unit multiplier $\mu = 1$, regardless of the structural constraints later placed on S.

Theorem 6.5 (The penalised objective selects an extremal-volume candidate)**.** *Let* $\mathcal{G} = \{g_{\eta,S} : S \in \mathcal{S}^d_{++}\}$.

(a) Upper CBO. *The problem* $\min_{\eta,S} \log \mathrm{vol}_\alpha(S)$ *subject to* $\overline{\mathrm{CBO}}(g_{\eta,S}) \le \log Z_{\max}$*, equivalently* $g_{\eta,S} \ge g^\star$ *pointwise (the smallest-volume candidate dominating* $g^\star$ *everywhere), has Lagrangian, at multiplier* $\mu = 1$ *and up to additive constants, exactly* $\mathrm{PCBO}_s(\eta, S)$ *of* (10) *with* $s = +1$.

(b) Lower CBO. *For a target level* $c \le \log Z_{\max}$*, the problem* $\max_{\eta,S} \log \mathrm{vol}_\alpha(S)$ *subject to* $\underline{\mathrm{CBO}}(g_{\eta,S}) \ge c$ *(at* $c = \log Z_{\max}$*, equivalently* $g_{\eta,S} \le g^\star$ *pointwise, the largest-volume candidate staying below* $g^\star$ *everywhere) has Lagrangian, at multiplier* $\mu = 1$ *and up to additive constants, exactly* $\mathrm{PCBO}_s(\eta, S)$ *with* $s = -1$.

Proof. (a) For the minimization problem, the standard Lagrangian is $L(\eta, S, \mu) = \log \mathrm{vol}_\alpha(S) + \mu(\overline{\mathrm{CBO}}(g_{\eta,S}) - \log Z_{\max})$. At $\mu = 1$, using (9) and dropping the additive constants $c(\alpha, d) - \log Z_{\max}$, this yields $\overline{\mathrm{CBO}}(g_{\eta,S}) - \frac{1}{2}\log\det S$, which is $\sup_\theta\{-\hat{\ell}(\theta) + \frac{1}{2}(\theta-\eta)^\top S(\theta-\eta)\} - \frac{1}{2}\log\det S$, matching (10) at $s = +1$.

(b) For the maximization problem, the standard Lagrangian is $L(\eta, S, \mu) = \log \mathrm{vol}_\alpha(S) + \mu(\underline{\mathrm{CBO}}(g_{\eta,S}) - c)$. At $\mu = 1$, substituting (9) and discarding additive constants similarly yields $\underline{\mathrm{CBO}}(g_{\eta,S}) - \frac{1}{2}\log\det S$, which is $\inf_\theta\{-\hat{\ell}(\theta) + \frac{1}{2}(\theta-\eta)^\top S(\theta-\eta)\} - \frac{1}{2}\log\det S$, matching (10) at $s = -1$. □

The penalty repairs the degeneracy. We verify this repair for the isotropic L^2-prior case of Lemma 6.3. Whether it also repairs the diagonal, LRD, and BD structures is not checked: diagonal domains have multivariate boundaries, LRD penalisation couples two logarithmic terms with $\|u_s^{\mathrm{LRD}}\|^2$ (Table 7), and BD replicates these couplings across blocks, so the scalar divergence argument does not transfer directly. We leave these open.

Geometric justification alone does not guarantee an interior stationary point, which requires the penalty to break the strict monotonicity of Lemma 6.3. Here, we relax the local confinement $\theta_s(\vartheta) \in U$ of Assumption 6.1(d) to $\mathbb{R}^d$. Because ℓ is exactly quadratic on $\mathbb{R}^d$, the inner objective is globally strictly concave ($s = +1$) or convex ($s = -1$) wherever $H_s(\vartheta) \succ 0$, so $\theta_s(\vartheta)$ is globally well-defined up to the spectral boundary where it leaves any bounded set. Proposition 6.7 shows that any $\zeta > 0$ breaks monotonicity to force an interior stationary point, which under this quadratic structure is also a global extremum.

Lemma 6.6 (Divergence of u_s at the positivity boundary, exact-quadratic case)**.** *Suppose* ℓ *is exactly quadratic on* $\mathbb{R}^d$*, as in Section 7.3, so* $\nabla^2\hat{\ell}$ *is constant and* $u_s(\vartheta) = -H_s(\vartheta)^{-1}\nabla\hat{\ell}(\eta)$ *holds exactly. Let* $\nabla^2\hat{\ell} = \sum_i \lambda_i v_i v_i^\top$ *be an orthonormal eigendecomposition, and assume* $\nabla\hat{\ell}(\eta)$ *has a nonzero component along an eigenvector attaining* $\lambda_{\min}(\nabla^2\hat{\ell})$ *for* $s = +1$ *(resp.* $\lambda_{\max}(\nabla^2\hat{\ell})$ *for* $s = -1$*). Then* $\|u_s(\vartheta)\| \to \infty$ *as* $\vartheta \to \lambda_{\min}(\nabla^2\hat{\ell})^-$ *for* $s = +1$*, and as* $\vartheta \to \lambda_{\max}(\nabla^2\hat{\ell})^+$ *for* $s = -1$.

Proof. For $s = +1$, $H_s(\vartheta) = \sum_i (\lambda_i - \vartheta) v_i v_i^\top \succ 0$ for $\vartheta < \lambda_{\min}(\nabla^2 \hat{\ell})$, so $u_{+1}(\vartheta) = -\sum_i (\lambda_i - \vartheta)^{-1} (v_i^\top \nabla \hat{\ell}(\eta)) v_i$. By orthogonality, $\|u_{+1}(\vartheta)\|^2 = \sum_i (\lambda_i - \vartheta)^{-2} (v_i^\top \nabla \hat{\ell}(\eta))^2$. The term attaining $\lambda_i = \lambda_{\min}(\nabla^2 \hat{\ell})$ with $v_i^\top \nabla \hat{\ell}(\eta) \neq 0$ diverges as $\vartheta \to \lambda_{\min}(\nabla^2 \hat{\ell})^-$, hence $\|u_{+1}(\vartheta)\| \to \infty$. The case $s = -1$ is identical with $H_s(\vartheta) = \sum_i (\vartheta - \lambda_i) v_i v_i^\top$, $u_{-1}(\vartheta) = -\sum_i (\vartheta - \lambda_i)^{-1} (v_i^\top \nabla \hat{\ell}(\eta)) v_i$, and $\vartheta \to \lambda_{\max}(\nabla^2 \hat{\ell})^+$. □

Proposition 6.7 (The penalty breaks monotonicity, exact-quadratic case)**.** *Fix any $\zeta > 0$. Suppose ℓ is exactly quadratic on $\mathbb{R}^d$ (so that Lemma 6.6 applies), the nonzero-component condition of Lemma 6.6 holds, and $\lambda_{\min}(\nabla^2 \hat{\ell}) > 0$ if $s = +1$ (resp. $\lambda_{\max}(\nabla^2 \hat{\ell}) > 0$ if $s = -1$). Under Assumption 6.1(a) and (b), $\mathrm{PCBO}_s(\eta, \cdot)$ has an interior stationary point $\vartheta^\star$ on its curvature-admissible domain $\{\vartheta > 0 : H_s(\vartheta) \succ 0\}$, which is a global minimum for $s = +1$ and a global maximum for $s = -1$.*

Proof. For any ϑ such that $H_s(\vartheta) \succ 0$, the inner objective is strictly concave on $\mathbb{R}^d$ for $s = +1$ (resp. strictly convex for $s = -1$), so its extremiser $\theta_s(\vartheta)$ is unique and interior to $\mathbb{R}^d$. By Danskin's theorem applied to the inner supremum ($s = +1$) and infimum ($s = -1$), $\mathrm{PCBO}_s(\eta, \cdot)$ is differentiable on $\{\vartheta > 0 : H_s(\vartheta) \succ 0\}$, with $\partial_\vartheta \mathrm{PCBO}_s = \frac{1}{2}\|u_s(\vartheta)\|^2 - \zeta/\vartheta$.

For $s = +1$, $\lambda_{\min}(\nabla^2 \hat{\ell}) > 0$ and $H_s(\vartheta) \succ 0$ defines the domain $(0, \lambda_{\min}(\nabla^2 \hat{\ell}))$. As $\vartheta \to 0^+$, $u_s(\vartheta) \to -[\nabla^2 \hat{\ell}]^{-1} \nabla \hat{\ell}(\eta)$ remains finite, so $-\zeta \log \vartheta \to +\infty$ dominates and $\mathrm{PCBO}_s \to +\infty$. As $\vartheta \to \lambda_{\min}(\nabla^2 \hat{\ell})^-$, Lemma 6.6 gives $\|u_s(\vartheta)\| \to \infty$, so the inner supremum diverges to $+\infty$ while $-\zeta \log \vartheta$ stays finite, and $\mathrm{PCBO}_s \to +\infty$ there too. Extending $\mathrm{PCBO}_s(\eta, \cdot)$ to $[0, \lambda_{\min}(\nabla^2 \hat{\ell})]$ by $+\infty$ at both endpoints makes it lower semicontinuous on a compact interval, hence attaining its infimum by the extreme value theorem, necessarily at an interior $\vartheta^\star$ since PCBO_s is finite somewhere inside, a global minimum satisfying $\partial_\vartheta \mathrm{PCBO}_s(\vartheta^\star) = 0$ by Fermat's theorem.

For $s = -1$, let $M \coloneqq \max(0, \lambda_{\max}(\nabla^2 \hat{\ell}))$, which coincides with $\lambda_{\max}(\nabla^2 \hat{\ell})$ throughout this report. The domain is (M, ∞). As $\vartheta \to M^+$, Lemma 6.6 gives $\|u_s(\vartheta)\| \to \infty$, so the inner infimum diverges to $-\infty$. As $\vartheta \to \infty$, $u_s(\vartheta) = \sum_i (\vartheta - \lambda_i)^{-1} (v_i^\top \nabla \hat{\ell}(\eta)) v_i = O(1/\vartheta) \to 0$, so the inner infimum stays bounded while $-\zeta \log \vartheta \to -\infty$ dominates, giving $\mathrm{PCBO}_s \to -\infty$ there too. The same argument, with $-\infty$ at both endpoints and upper semicontinuity, gives an interior global maximum $\vartheta^\star$ satisfying $\partial_\vartheta \mathrm{PCBO}_s(\vartheta^\star) = 0$. □

Remark 6.8 (Scope of the global-extremum claim)*.* Existence of an interior stationary point does not rely on the exact-quadratic hypothesis: under Assumption 6.1, the first-order balance $\frac{1}{2}\|u_s(\vartheta)\|^2 = \zeta/\vartheta$ holds locally. Proving it is a global extremum does, since quadraticity gives $u_s(\vartheta)$ an explicit pole at the spectral boundary (Lemma 6.6), driving PCBO_s to $\pm\infty$ at both ends of the domain. For general losses $\|u_s(\vartheta)\|$ is not guaranteed to diverge there, so whether $\vartheta^\star$ remains a global extremum is left open.

Proposition 6.7 thus breaks the strict monotonicity of Lemma 6.3 for both signs and any $\zeta > 0$, at least in the exact-quadratic case: the interior stationary point that Section 6.2 (Proposition 6.10) characterises once $\zeta = d/2$ is fixed below is a global minimum for $s = +1$ and a global maximum for $s = -1$ there.

Closed-form penalties. For the LRD candidate $S^{-1} = u_s^{\mathrm{LRD}} (u_s^{\mathrm{LRD}})^\top + \psi I$ (justified in Section 6.3), the eigenvalues of S are $1/\psi$ (multiplicity $d - 1$) and $1/(\psi + \|u_s^{\mathrm{LRD}}\|^2)$ by Lemma 6.19, which yields $-\frac{1}{2} \log\det S = \frac{1}{2}(d-1) \log \psi + \frac{1}{2} \log(\psi + \|u_s^{\mathrm{LRD}}\|^2)$. For block-diagonal $S = \bigoplus_\ell S_\ell$ with LRD blocks $S_\ell^{-1} = u_{s,\ell}^{\mathrm{BD}} (u_{s,\ell}^{\mathrm{BD}})^\top + \psi_\ell I$, the spectrum is the union of each block's, so $-\frac{1}{2} \log\det S = \sum_\ell \big(\frac{1}{2}(d_\ell - 1) \log \psi_\ell + \frac{1}{2} \log(\psi_\ell + \|u_{s,\ell}^{\mathrm{BD}}\|^2)\big)$. For isotropic and diagonal precisions, the determinants are computed directly.

Proposition 6.9 (Restrictions of the log-determinant penalty)**.** *For $S = \vartheta I$, $-\frac{1}{2} \log\det S = -\frac{d}{2} \log \vartheta$, and for $S = \mathrm{diag}(\boldsymbol{\vartheta})$, $-\frac{1}{2} \log\det S = -\frac{1}{2} \sum_{j=1}^d \log \vartheta_j$. In particular, $-\zeta \log \vartheta$ is the isotropic restriction of the matrix penalty only for $\zeta = d/2$.*

Proof. Immediate from $\log\det(\vartheta I) = d \log \vartheta$ and $\log\det \mathrm{diag}(\boldsymbol{\vartheta}) = \sum_j \log \vartheta_j$. The last statement follows since $-\zeta \log \vartheta = -\frac{d}{2} \log \vartheta$ for all $\vartheta > 0$ forces $\zeta = d/2$. □

Table 7: The specificity penalty $-\frac{1}{2}\log\det S$ of Theorem 6.5, evaluated for each precision structure used in this report.

Structure	Precision S	Penalty $-\frac{1}{2}\log\det S$
Isotropic	ϑI	$-\frac{d}{2}\log\vartheta$
Diagonal	$\mathrm{diag}(\boldsymbol{\vartheta})$	$-\frac{1}{2}\sum_{j=1}^{d}\log\vartheta_j$
LRD	$(u_s^{\mathrm{LRD}}(u_s^{\mathrm{LRD}})^\top + \psi I)^{-1}$	$\frac{1}{2}(d-1)\log\psi + \frac{1}{2}\log(\psi + \|u_s^{\mathrm{LRD}}\|^2)$
Block-diagonal	$\bigoplus_\ell S_\ell$	$\sum_\ell\big(\frac{1}{2}(d_\ell-1)\log\psi_\ell + \frac{1}{2}\log(\psi_\ell + \|u_{s,\ell}^{\mathrm{BD}}\|^2)\big)$

6.2 Isotropic and diagonal precisions: a damped Picard fixed point

Stationarity condition. The penalty of §6.1 breaks the monotonicity of CBO_s in the precision, so $\partial_\vartheta \mathrm{PCBO}_s = 0$ now has an interior solution. This subsection treats the isotropic and diagonal cases, extending Assumption 6.1 to $\boldsymbol{\vartheta} \in \mathbb{R}^d_{>0}$. Parts (a) and (c) carry over unchanged. Part (b) becomes $H_s^{\mathrm{diag}} = s(\nabla^2\hat{\ell}(\eta) - \mathrm{diag}(\boldsymbol{\vartheta})) \succ 0$. Part (d) becomes, at fixed η, that $\theta_s^{\mathrm{diag}}(\boldsymbol{\vartheta})$ exists, is unique, lies in U with $(\theta_s^{\mathrm{diag}} - \eta)_j \neq 0$ for all j, and is continuous in $\boldsymbol{\vartheta}$.

Proposition 6.10 (Isotropic and diagonal stationarity and second-order approximation)**.** *Under Assumption 6.1, the exact stationarity conditions $\partial_\vartheta \mathrm{PCBO}_s = 0$ (isotropic) and $\partial_{\vartheta_j}\mathrm{PCBO}_s = 0$ (diagonal) read*

$$\vartheta^\star = \frac{d}{\|\theta_s - \eta\|^2}, \qquad \vartheta_j^\star = \frac{1}{(\theta_s^{\mathrm{diag}} - \eta)_j^2}. \tag{11}$$

Approximating the displacement $\theta_s^X - \eta \approx s\,u_s^X$ with $u_s^X := -H_s^{X-1}\nabla\hat{\ell}(\eta)$ via the second-order Taylor expansion around η, yields $(\theta_s^X - \eta)_j^2 \approx (u_s^X)_j^2$ and the operational implicit fixed-point equations

$$\vartheta^\star \approx \frac{d}{\|u_s\|^2}, \qquad \vartheta_j^\star \approx \frac{1}{(u_s^{\mathrm{diag}})_j^2}, \tag{12}$$

with $H_s = s(\nabla^2\hat{\ell}(\eta) - \vartheta I)$ and $H_s^{\mathrm{diag}} = s(\nabla^2\hat{\ell}(\eta) - \mathrm{diag}(\boldsymbol{\vartheta}))$, implicit since H_s and H_s^{diag} depend on the precision.

Proof. By Assumption 6.1, each inner extremiser is unique and interior, so Danskin's theorem applies. For the isotropic candidate, $\partial_\vartheta \mathrm{PCBO}_s = \frac{1}{2}\|\theta_s - \eta\|^2 - \frac{d}{2\vartheta}$. For the diagonal candidate, $\partial_{\vartheta_j}\mathrm{PCBO}_s = \frac{1}{2}(\theta_s^{\mathrm{diag}} - \eta)_j^2 - \frac{1}{2\vartheta_j}$. Setting these to zero yields (11). Substituting the second-order displacement $\theta_s^X - \eta \approx s\,u_s^X$ with $u_s^X = -H_s^{X-1}\nabla\hat{\ell}(\eta)$ yields $(\theta_s^X - \eta)_j^2 \approx (u_s^X)_j^2$ since $s^2 = 1$, giving (12) with $u_s = -s(\nabla^2\hat{\ell}(\eta) - \vartheta I)^{-1}\nabla\hat{\ell}(\eta)$ and $u_s^{\mathrm{diag}} = -s(\nabla^2\hat{\ell}(\eta) - \mathrm{diag}(\boldsymbol{\vartheta}))^{-1}\nabla\hat{\ell}(\eta)$, the forms implemented in Algorithms 1 and 2. □

Remark 6.11 (When the diagonal condition decouples coordinatewise). Equation (12) holds in general: $\vartheta_j^\star$ is set by coordinate j alone whenever $\nabla^2\hat{\ell}(\eta)$ is diagonal, or replaced by the diagonal proxy $h \approx \nabla^2\ell(\eta)$ (§7.3), giving $H_s^{\mathrm{diag}} \approx s\,\mathrm{diag}(h + \delta - \boldsymbol{\vartheta})$. Otherwise, inverting H_s^{diag} couples all coordinates into $(u_s^{\mathrm{diag}})_j$.

Remark 6.12 (Relation to Vprop/IVON). Probabilistically, the optimal diagonal precision obeys $\vartheta_j^\star \approx \mathbb{E}_q[\nabla^2_{jj}\ell] + \delta$ by Price's theorem [24], as in Vprop [25] and IVON [8]. The possibilistic condition $\vartheta_j^\star = 1/(u_s^{\mathrm{diag}})_j^2$ instead sets precision from the squared Newton step, providing a structurally distinct estimator (§7.3).

Remark 6.13 (Why Picard for ϑ). Solving the implicit fixed point $\vartheta = d/\|u_s(\vartheta)\|^2$ via Picard iteration offers three practical advantages over Newton's method: (i) lower cost, reusing the vector u_s computed for the η-update without evaluating Φ', (ii) robustness, avoiding division by $1 - \Phi'$ and keeping the capped step $\mathcal{C}_s(\Phi(\vartheta_t))$ strictly confined to $\mathcal{I}_s$ (15), and (iii) simpler convergence guarantees (Theorem 6.14), requiring only a bound on $|\Phi'|$ rather than Lipschitz continuity of Φ'.

The damped Picard map. Under this proxy, $H_s \approx s(\mathrm{diag}(h+\delta) - \vartheta I)$ has eigenvalues $s(h_j+\delta-\vartheta)$, positive iff $\vartheta < h_j + \delta$ for $s = +1$ or $\vartheta > h_j + \delta$ for $s = -1$: $h_j + \delta$ is the boundary of $H_s \succ 0$ for coordinate j. For the isotropic candidate this becomes $\min_j(h_j + \delta)$ (resp. $\max_j$), coinciding with the diagonal boundary $h_0 + \delta$ at initialisation, where $h_{j,0} = h_0$ uniformly. Because the right-hand side of (12) depends on ϑ through H_s, we solve this implicit fixed point via the damped Picard map

$$T_{w_t}(\vartheta_t) \coloneqq (1 - w_t)\vartheta_t + w_t\, \mathcal{C}_s\big(\Phi(\vartheta_t)\big) \qquad (\text{resp. } T^{\mathrm{diag}}_{w_t}(\boldsymbol{\vartheta}_t)_j \coloneqq (1 - w_t)\vartheta_{j,t} + w_t\, \mathcal{C}^{\mathrm{diag}}_s\big(\Phi^{\mathrm{diag}}_j(\boldsymbol{\vartheta}_t)\big)), \tag{13}$$

where $\Phi(\vartheta) \coloneqq d/\|u_s\|^2$, $\Phi^{\mathrm{diag}}_j(\boldsymbol{\vartheta}) \coloneqq 1/(u^{\mathrm{diag}}_s)^2_j$, and $\mathcal{C}_s(\cdot)$, $\mathcal{C}^{\mathrm{diag}}_s(\cdot)$, defined below in (14), are applied before mixing with the damping weight w_t, a Krasnoselskii–Mann-type relaxation [26].

Setting $w_t = 1$ recovers the undamped iteration. Iterates are initialised at $\vartheta_0 = (h_0 + \delta)(1 \mp \epsilon_{\mathrm{rel}})$ for $s = \pm 1$ (resp. $\vartheta_{j,0} = (h_{j,0} + \delta)(1 \mp \epsilon_{\mathrm{rel}})$), where $\epsilon_{\mathrm{rel}} > 0$ places the starting point symmetrically close to the boundary. This renders Proposition 6.10's implicit condition operational, with damping stabilising the trajectory when Φ (resp. Φ^{diag}_j) varies rapidly.

The anti-collapse and positivity cap. Enforcing positive definiteness on a diagonal variational precision by an explicit, closed-form bound is a known device in natural-gradient VI, where naive updates do not guarantee it [27]. We define $\mathcal{C}^{\mathrm{diag}}_s(\cdot)$ to combine two constraints: preventing Dirac collapse and ensuring $H_s \succ 0$. For $s = +1$, a lower floor $v_{\min} > 0$ is added to bound ϑ_j on both sides for either sign:

$$\begin{aligned}
\mathcal{C}^{\mathrm{diag}}_{+1}(\vartheta_j) &\coloneqq \max\Big(\min\big(\vartheta_j,\ (h_j + \delta)/r,\ (h_j + \delta)(1 - \epsilon_{\mathrm{rel}})\big),\ v_{\min}\Big),\\
\mathcal{C}^{\mathrm{diag}}_{-1}(\vartheta_j) &\coloneqq \max\big(\min(\vartheta_j,\ (h_j + \delta)/r),\ (h_j + \delta)(1 + \epsilon_{\mathrm{rel}})\big),
\end{aligned} \tag{14}$$

where $\mathcal{C}^{\mathrm{diag}}_s$ defines non-empty intervals whenever $v_{\min} < (h_j+\delta)(1-\epsilon_{\mathrm{rel}})$ for $s = +1$, and $r < 1/(1+\epsilon_{\mathrm{rel}})$ for $s = -1$. The isotropic cap $\mathcal{C}_s(\vartheta)$ follows the same expressions with $\vartheta_j \to \vartheta$, $h_j + \delta \to \min_j(h_j + \delta)$ $(s = +1)$ or $\max_j(h_j + \delta)$ $(s = -1)$, and a scalar $v_{\min}$.

The anti-collapse bound $(h_j+\delta)/r$ is common to both signs, as $\vartheta^\star_j \to \infty$ when $u^{\mathrm{diag}}_s \to 0$: it enforces $(h_j + \delta)/\vartheta_j \geq r$, lower-bounding the product of variance and curvature. Approaching the boundary $h_j + \delta$ drives $\|u_s(\vartheta)\| \to \infty$ (Lemma 6.6), so the margin ϵ_{rel} bounds ϑ_j away from it. For $s = +1$, one can choose $r \ll 1$, leaving the positivity bound $\vartheta_j \leq (h_j + \delta)(1 - \epsilon_{\mathrm{rel}})$ active. For $s = -1$, $H_s \succ 0$ requires ϑ_j above $h_j + \delta$, pulling opposite to anti-collapse. Here $r \ll 1$ is prohibited: $\vartheta^\star_j \approx 1/u^2_j$ easily scales above the curvature, saturating $\min(\vartheta^\star_j, (h_j+\delta)/r)$. Choosing $r \ll 1$ would drive $\vartheta_j \propto 1/r \to \infty$, blowing up the denominator $\vartheta_j - (h_j + \delta) \approx (h_j + \delta)/r$ and sending the update $u_s \propto r$ to zero, thereby freezing optimisation.

Table 8: Constraints on ϑ_j (diagonal) or ϑ (isotropic, replacing $h_j + \delta$ with $\min_j$ or $\max_j$).

	$s = +1$	$s = -1$
Anti-collapse	$\vartheta_j \leq (h_j + \delta)/r$	$\vartheta_j \leq (h_j + \delta)/r$
Positivity	$\vartheta_j \leq (h_j + \delta)(1 - \epsilon_{\mathrm{rel}})$	$\vartheta_j \geq (h_j + \delta)(1 + \epsilon_{\mathrm{rel}})$
Floor	$\vartheta_j \geq v_{\min}$ (rarely active)	—
Active bound	positivity	opposing pair

Confining the iterates to U. Assumption 6.1(a) provides an open neighbourhood U of η on which ℓ is C^2, containing an open Euclidean ball $B(\eta, R_U) \subseteq U$. Condition (d) holds if $\theta_s(\vartheta) \in B(\eta, R_U)$ for every capped ϑ. By construction, the cap enforces $\lambda_{\min}(H_s(\vartheta)) \geq \epsilon_{\mathrm{rel}} \min_j(h_j + \delta)$ for both signs, yielding $\|H_s(\vartheta)^{-1}\|_2 \leq 1/(\epsilon_{\mathrm{rel}} \min_j(h_j + \delta))$ uniformly. Under the second-order step, this implies $\|u_s(\vartheta)\| \leq \|\nabla\hat{\ell}(\eta)\|/(\epsilon_{\mathrm{rel}} \min_j(h_j + \delta))$. Choosing $\epsilon_{\mathrm{rel}} \geq \|\nabla\hat{\ell}(\eta)\|/(R_U \min_j(h_j + \delta))$ thus ensures $\theta_s(\vartheta) \in U$ throughout the inner Picard iteration at this η. This proves local satisfiability of (d); it does not guarantee global smoothness of ℓ, nor that a single ϵ_{rel} holds across outer mode updates, as U, R_U, and curvature estimates change during training.

The compact interval $\mathcal{I}_s$. The cap bounds ϑ_j on both sides for either sign:

$$\mathcal{I}_{s,j} \coloneqq \begin{cases} \big[v_{\min},\ (h_j+\delta)(1-\epsilon_{\mathrm{rel}})\big] & s=+1, \\ \big[(h_j+\delta)(1+\epsilon_{\mathrm{rel}}),\ (h_j+\delta)/r\big] & s=-1, \end{cases} \tag{15}$$

nonempty whenever $v_{\min} < (h_j+\delta)(1-\epsilon_{\mathrm{rel}})$ $(s=+1)$ and $r < 1/(1+\epsilon_{\mathrm{rel}})$ $(s=-1)$. Both bounds tighten as ϵ_{rel} grows, which scales inversely with the neighborhood radius R_U. In practice, taking $v_{\min}$ and r safely small guarantees non-emptiness. The isotropic interval $\mathcal{I}_s$ follows the same scalar correspondence.

Conditions for convergence. Recall $\Phi(\vartheta) \coloneqq d/\|u_s(\vartheta)\|^2$ from (13), with $u_s(\vartheta) = -H_s(\vartheta)^{-1}\nabla\hat{\ell}(\eta)$ at fixed η. By the quotient rule, $\Phi'(\vartheta) = -2d\,(u_s)^\top \partial_\vartheta u_s/\|u_s(\vartheta)\|^4$, which remains bounded on $\mathcal{I}_s$ since $\|u_s\|$ is bounded away from 0 (Assumption 6.1(d)) and u_s, $\partial_\vartheta u_s$ stay bounded. This reduces to bounding H_s^{-1} and $\partial_\vartheta(H_s^{-1}) = sH_s^{-2}$ (differentiating $H_s^{-1}H_s = I$ with $\partial_\vartheta H_s = -sI$). Because $H_s(\vartheta) = s(\nabla^2\hat{\ell}(\eta) - \vartheta I)$ is affine, $\vartheta \mapsto \lambda_{\min}(H_s(\vartheta))$ is continuous and strictly positive on $\mathcal{I}_s$, hence attaining a positive infimum on this compact set. This yields uniform bounds $\|H_s^{-1}\|_2 = 1/\lambda_{\min}(H_s)$ and $\|\partial_\vartheta(H_s^{-1})\|_2 = 1/\lambda_{\min}(H_s)^2$. Consequently, a finite Lipschitz constant $|\Phi'| \le L_\Phi$ (and $|\Phi_j^{\mathrm{diag}\prime}| \le L_{\Phi,j}$) always exists on $\mathcal{I}_s$, though $L_\Phi < 1$ is not automatic and must be verified.

Theorem 6.14 (Sufficient condition for linear convergence on an invariant interval)**.** *Let Assumption 6.1 hold, $\mathcal{I}_s \subset (0,\infty)$ a closed interval mapped into itself by T_w for some $w \in (0,1]$, and $|\Phi'(\vartheta)| \le L_\Phi$ on $\mathcal{I}_s$. Setting $\kappa(w) \coloneqq 1 + w(L_\Phi - 1)$, T_w is $\kappa(w)$-Lipschitz on $\mathcal{I}_s$. If $L_\Phi < 1$, then $\kappa(w) < 1$ for every $w \in (0,1]$, making T_w a strict contraction with a unique fixed point $\vartheta^\dagger \in \mathcal{I}_s$ satisfying $|\vartheta_t - \vartheta^\dagger| \le \kappa(w)^t|\vartheta_0 - \vartheta^\dagger|$. The same applies coordinatewise on $\prod_j \mathcal{I}_{s,j}$ once $|\Phi_j^{\mathrm{diag}\prime}| \le L_{\Phi,j} < 1$. This condition is sufficient, discarding the sign of Φ' (Rem. 6.16).*

Proof. Because the projection operator $\mathcal{C}_s$ onto the closed convex interval $\mathcal{I}_s$ is non-expansive (1-Lipschitz), $|\mathcal{C}_s(x) - \mathcal{C}_s(y)| \le |x-y|$. By the mean value theorem, $|\Phi(\vartheta) - \Phi(\vartheta')| \le L_\Phi|\vartheta - \vartheta'|$ on $\mathcal{I}_s$. Decomposing $T_w(\vartheta) - T_w(\vartheta') = (1-w)(\vartheta-\vartheta') + w(\mathcal{C}_s(\Phi(\vartheta)) - \mathcal{C}_s(\Phi(\vartheta')))$ with $1-w \ge 0$, the triangle inequality gives

$$|T_w(\vartheta) - T_w(\vartheta')| \le (1-w)|\vartheta-\vartheta'| + w|\mathcal{C}_s(\Phi(\vartheta)) - \mathcal{C}_s(\Phi(\vartheta'))| \le \kappa(w)|\vartheta-\vartheta'|.$$

If $L_\Phi < 1$, then $\kappa(w) < 1$, so T_w is a contraction on the complete metric space $\mathcal{I}_s$ and Banach's theorem applies. The coordinatewise extension with $L_{\Phi,j}$ on $\mathcal{I}_{s,j}$ follows identically. □

Remark 6.15 (Explicit L_Φ, and degradation near stationarity). With $g \coloneqq \nabla\hat{\ell}(\eta)$ and $a_j(\vartheta) \coloneqq h_j + \delta - \vartheta$, $\Phi'(\vartheta) = -\frac{2d}{\|u_s(\vartheta)\|^4}\sum_j \frac{g_j^2}{a_j(\vartheta)^3}$. For nonzero g, setting $u_{\min}(\eta) \coloneqq \min_{\mathcal{I}_s}\|u_s\| > 0$ and $a_{\min} \coloneqq \epsilon_{\mathrm{rel}}\min_j(h_j+\delta)$ yields $|\Phi'(\vartheta)| \le \frac{2d\|g\|^2}{u_{\min}(\eta)^4}\max_j|a_j(\vartheta)|^{-3} \le \frac{2d\|g\|^2}{u_{\min}(\eta)^4 a_{\min}^3} \coloneqq L_\Phi$. This bound degrades near stationary points ($\|u_s\| \propto \|g\| \to 0$) and is uninformative at network scale (§7.3), where L_Φ is evaluated numerically.

Remark 6.16 (When damping restores contraction, and why only for $s=+1$). Positivity of H_s forces $a_j(\vartheta) = h_j + \delta - \vartheta$ to keep a fixed sign on $\mathcal{I}_s$: positive for $s=+1$, negative for $s=-1$. By Rem. 6.15, Φ' inherits the opposite sign: strictly negative for $s=+1$ and strictly positive for $s=-1$. For $s=+1$, writing $\Phi' = -c(\vartheta)$ with $c(\vartheta) \in [0, L_\Phi]$, the exact derivative of T_w is $1 - w(1+c(\vartheta))$, which remains in $(-1,1)$ for $w < 2/(1+L_\Phi)$, restoring contraction even where $\kappa(w) \ge 1$. For $s=-1$, the derivative $1 + w(c(\vartheta)-1)$ exceeds 1 whenever $c(\vartheta) > 1$, so damping cannot repair expansion.

Remark 6.17 (Scope). Fixing η, Theorem 6.14 confines ϑ to a compact set and proves contraction under a checkable condition. Alternating decoupled updates leaves both simultaneous optimisation and joint (η,ϑ) convergence open, typically requiring two-timescale analysis [28].

6.3 General and block-diagonal precisions: rank deficiency and sign-asymmetric safeguards

We have treated the isotropic and diagonal precisions. We now consider a general symmetric positive-definite precision $S \in \mathcal{S}^d_{++}$, subsuming both as special cases, as well as a block-diagonal precision $S = \bigoplus_\ell S_\ell$. Throughout this subsection, $\preceq$ and $\succeq$ denote the Loewner partial order on symmetric matrices ($A \preceq B \iff B - A$ is positive semi-definite), not to be confused with the possibilistic specificity order $\preceq$ on $\mathcal{F}(\Theta)$. We extend Assumption 6.1 to S as in §6.2: parts (a) and (c) are unaffected, (b) becomes $H_s = s(\nabla^2\hat{\ell}(\eta) - S) \succ 0$, and (d) requires that, at fixed η, the inner extremiser $\theta^{\mathrm{LRD}}_s(S)$ exists, is unique, lies in U, and is continuous in S. For the block-diagonal precision of Theorem 6.24, (d) is assumed to hold blockwise for each local extremiser $\theta^{\mathrm{BD}}_{s,\ell}(S_\ell)$ under the block-separable surrogate.

However, this generality comes at a cost: the stationarity condition that uniquely determined $\vartheta^\star$ (isotropic) or $\vartheta^\star_j$ (diagonal) no longer yields an invertible precision for general S. Proposition 6.18 below identifies this rank deficiency, motivating a structured low-rank-plus-diagonal regularisation similar to low-rank covariance approximations in VI [29], while whether this specific form represents the optimal regularisation in our context remains an open question.

Proposition 6.18 (Singular stationarity and LRD regularisation)**.** *With the log-volume penalty $-\frac{1}{2}\log\det S$, stationarity of $\mathrm{PCBO}_s(\eta, S)$ with respect to $S \in \mathcal{S}^d_{++}$ under the Frobenius inner product $\langle A, B\rangle := \mathrm{tr}(A^\top B)$ requires*

$$\nabla_S \mathrm{PCBO}_s = \tfrac{1}{2} u_s^{\mathrm{LRD}} (u_s^{\mathrm{LRD}})^\top - \tfrac{1}{2} S^{-1} = 0, \qquad \text{so} \qquad S^{-1} = S_\star^{-1} := u_s^{\mathrm{LRD}} (u_s^{\mathrm{LRD}})^\top,$$

which has rank one and is therefore singular for $d > 1$. We regularise $S_\star^{-1}$ to the low-rank-plus-diagonal (LRD) family

$$S^{-1} = u_s^{\mathrm{LRD}} (u_s^{\mathrm{LRD}})^\top + \psi I, \qquad \psi > 0, \tag{16}$$

restoring $S \in \mathcal{S}^d_{++}$.

Proof. By Assumption 6.1(d) extended to S, $\theta^{\mathrm{LRD}}_s(S)$ exists, is unique, lies in U, and satisfies $\theta^{\mathrm{LRD}}_s(S) \neq \eta$, so Danskin's theorem applies to $\mathrm{PCBO}_s(\eta, \cdot)$ at S, giving, with $\theta^{\mathrm{LRD}}_s(S) - \eta \approx s\, u_s^{\mathrm{LRD}}$ and $s^2 = 1$,

$$\nabla_S \mathrm{PCBO}_s \approx \tfrac{1}{2} u_s^{\mathrm{LRD}} (u_s^{\mathrm{LRD}})^\top - \tfrac{1}{2} S^{-1}.$$

Setting $\nabla_S \mathrm{PCBO}_s = 0$ yields $S_\star^{-1} := u_s^{\mathrm{LRD}} (u_s^{\mathrm{LRD}})^\top$, a rank-one matrix (dimension 1 image since $u_s^{\mathrm{LRD}} \neq 0$ by (d)), hence singular for $d > 1$: no $S \in \mathcal{S}^d_{++}$ solves the stationarity condition exactly. Adding ψI to $S_\star^{-1}$, (16) defines $S^{-1} := S_\star^{-1} + \psi I$, the sum of a positive semi-definite and a strictly positive definite matrix, hence positive definite: for every $v \neq 0$, $v^\top S^{-1} v \geq \psi \|v\|^2 > 0$. Its inverse $S := (S^{-1})^{-1} \in \mathcal{S}^d_{++}$ therefore exists, as made explicit in Lemma 6.19. □

Lemma 6.19 (Spectrum of the LRD regulariser)**.** *For $S^{-1} = u_s^{\mathrm{LRD}} (u_s^{\mathrm{LRD}})^\top + \psi I$ with $\psi > 0$ and $u_s^{\mathrm{LRD}} \neq 0$, S^{-1} has eigenvalue $\psi + \|u_s^{\mathrm{LRD}}\|^2$ on $\mathrm{span}\{u_s^{\mathrm{LRD}}\}$ and ψ on $(u_s^{\mathrm{LRD}})^\perp$. By Sherman–Morrison,*

$$S = \tfrac{1}{\psi} I - \frac{u_s^{\mathrm{LRD}} (u_s^{\mathrm{LRD}})^\top}{\psi(\psi + \|u_s^{\mathrm{LRD}}\|^2)},$$

so S has reciprocal eigenvalues on the same eigenspaces: $\lambda_{\min}(S) = 1/(\psi + \|u_s^{\mathrm{LRD}}\|^2)$ on $\mathrm{span}\{u_s^{\mathrm{LRD}}\}$, and $\lambda_{\max}(S) = 1/\psi$ on $(u_s^{\mathrm{LRD}})^\perp$. In particular $S \preceq \frac{1}{\psi} I$, and every eigenvalue of S is strictly decreasing in ψ.

We now build explicit safeguards for positivity $H_s \succ 0$, in the same spirit as the caps of §6.2 and, more broadly, of the explicit constraints used to enforce positive definiteness in natural-gradient VI [27], seeking in each case the smallest reasonable ψ that certifies it. As in §6.2, h throughout this subsection denotes the diagonal curvature proxy estimating $\nabla^2\ell(\eta)$ (the EMA of squared stochastic gradients of §3.4). Write $m := \min_j(h_j + \delta)$, $M := \max_j(h_j + \delta)$.

Theorem 6.20 (Unconditional safeguard, $s = +1$)**.** *Let $s = +1$, so $H_s \approx \operatorname{diag}(h + \delta) - S$ and $H_s \succ 0 \iff S \prec \operatorname{diag}(h + \delta)$ under this proxy. A sufficient condition is $\lambda_{\max}(S) < m$, equivalently $\psi > 1/m$. The choice*

$$\psi = 1/m + a, \qquad a > 0,$$

satisfies it strictly, approaching the bound $1/m$ from above by a small margin a.

Proof. By Lemma 6.19, $\lambda_{\max}(S) = 1/\psi$, so $\lambda_{\max}(S) < m \iff \psi > 1/m$. If this holds, $S \preceq \lambda_{\max}(S)I \prec mI \preceq \operatorname{diag}(h + \delta)$, ensuring $H_s \succ 0$. Choosing $\psi = 1/m + a$ satisfies this lower bound by construction for any $a > 0$. The condition is sufficient but not necessary: $S \prec \operatorname{diag}(h + \delta)$ can hold with $\lambda_{\max}(S) \geq m$ if the maximal eigenvalue of S aligns with large coordinates of $\operatorname{diag}(h + \delta)$. □

Theorem 6.21 (Conditional safeguard, $s = -1$)**.** *Let $s = -1$, so $H_s \approx S - \operatorname{diag}(h + \delta)$ and $H_s \succ 0 \iff S \succ \operatorname{diag}(h + \delta)$ under this proxy. A sufficient condition is $\lambda_{\min}(S) > M$, equivalently $\psi + \|u_s^{\mathrm{LRD}}\|^2 < 1/M$, an upper bound on ψ, unlike Thm. 6.20. Since this bound moves with M and $\|u_s^{\mathrm{LRD}}\|^2$ at each step, we use multiplicative damping [30] rather than an additive offset:*

$$\psi = \frac{1}{\varphi}\Big(\tfrac{1}{M} - \|u_s^{\mathrm{LRD}}\|^2\Big), \qquad \varphi > 1,$$

which approaches $1/M - \|u_s^{\mathrm{LRD}}\|^2$ from below by the scaling factor $1/\varphi \in (0, 1)$, valid if and only if

$$\|u_s^{\mathrm{LRD}}\|^2 < \frac{1}{M} = \frac{1}{\max_j(h_j + \delta)}. \tag{17}$$

Proof. By Lemma 6.19, $\lambda_{\min}(S) = 1/(\psi + \|u_s^{\mathrm{LRD}}\|^2)$, so $\lambda_{\min}(S) > M \iff \psi < 1/M - \|u_s^{\mathrm{LRD}}\|^2$. If this holds, $S \succeq \lambda_{\min}(S)I \succ MI \succeq \operatorname{diag}(h + \delta)$, giving $H_s \succ 0$ under the proxy. This upper bound is strictly positive if and only if $\|u_s^{\mathrm{LRD}}\|^2 < 1/M$, which is (17). Under (17), $\psi = \varphi^{-1}(1/M - \|u_s^{\mathrm{LRD}}\|^2)$ guarantees both $H_s \succ 0$ and $\psi > 0$ (hence $S^{-1} \succ 0$) for any $\varphi > 1$. □

Remark 6.22 (The $s = -1$ safeguard is only conditional). Unlike Thm. 6.20, the safeguard of Thm. 6.21 applies only under (17): when $\|u_s^{\mathrm{LRD}}\|^2 \geq 1/M$, no $\psi > 0$ is feasible, which occurs early in training when the Newton step is large. A natural remedy is to run with $s = +1$ until $\|u_s^{\mathrm{LRD}}\|^2$ drops below $u_{\max}^2 := (1 - \xi)/M$ with $\xi \in (0, 1)$, then switch to $s = -1$. Setting $u_{\max}^2$ arbitrarily close to $1/M$ would leave ψ near 0, inducing a needle-like degeneracy (Rem. 6.23); the margin ξ guarantees available headroom of at least ξ/M.

Remark 6.23 (Explicit ψ, no separate collapse safeguard needed). The LRD candidate exhibits an analogue of the isotropic/diagonal collapse: by Lemma 6.19, $\psi \to 0^+$ yields $\lambda_{\max}(S) \to \infty$ while $\lambda_{\min}(S) \to 1/\|u_s^{\mathrm{LRD}}\|^2$ stays finite (variance vanishes orthogonally to u_s^{LRD} but remains bounded along it), whereas $\psi \to \infty$ gives $S \to 0$. Unlike the iterated target $\vartheta^\star = d/\|u_s\|^2$, ψ here is computed in closed form without Picard iterations: set directly from m for $s = +1$, and scaled from $1/M - \|u_s^{\mathrm{LRD}}\|^2$ by $1/\varphi$ for $s = -1$. Neither requires a separate cap. Since $m, M \geq \delta > 0$, we have $\psi = 1/m + a \in (a, 1/\delta + a]$ for $s = +1$, and $\psi \in [\xi/(\varphi M), 1/(\varphi\delta)]$ for $s = -1$, both strictly bounded away from 0 and ∞.

Theorem 6.24 (Block safeguards, both signs)**.** *Partition $\theta = (\theta_1, \ldots, \theta_L)$ into L blocks and let $S = \bigoplus_\ell S_\ell$ with $S_\ell^{-1} = u_{s,\ell}^{\mathrm{BD}}(u_{s,\ell}^{\mathrm{BD}})^\top + \psi_\ell I_{d_\ell}$, $m_\ell = \min_{j\in\ell}(h_j + \delta)$, and $M_\ell = \max_{j\in\ell}(h_j + \delta)$. Since a symmetric block-diagonal matrix is positive definite iff each block is, $H_s \succ 0$ reduces to per-block conditions: (a) for $s = +1$, $\psi_\ell = 1/m_\ell + a$ suffices for every block, and (b) for $s = -1$, $\psi_\ell = \varphi^{-1}(1/M_\ell - \|u_{s,\ell}^{\mathrm{BD}}\|^2)$ suffices whenever $\|u_{s,\ell}^{\mathrm{BD}}\|^2 < 1/M_\ell$. Blocks violating this condition run with $s = +1$ instead until $\|u_{s,\ell}^{\mathrm{BD}}\|^2$ drops below the threshold $u_{\max,\ell}^2 := (1 - \xi)/M_\ell$ with $\xi \in (0, 1)$ (Rem. 6.22).*

Proof. Under the proxy, H_s is block-diagonal with diagonal blocks $\operatorname{diag}(h_\ell + \delta) - S_\ell$ for $s = +1$, and $S_\ell - \operatorname{diag}(h_\ell + \delta)$ for $s = -1$. Because positive definiteness holds globally iff it holds on each diagonal block, claims (a) and (b) follow directly by applying Theorems 6.20 and 6.21 to block ℓ in isolation, replacing $(m, M, u_s^{\mathrm{LRD}})$ with $(m_\ell, M_\ell, u_{s,\ell}^{\mathrm{BD}})$. □

7 Applying and Validating the Construction

Having constructed the learned precision in Section 6, this section puts it to work. §7.1 connects the lower CBO to Sharpness-Aware Minimisation, replacing SAM's hand-tuned radius by one derived automatically from the learned precision. §7.2 assembles the constructions of Section 6 into eight named optimisers and analyses their complexity. §7.3 then tests all of this empirically, on a toy problem where every quantity is exact, and on LeNet-5/Fashion-MNIST.

7.1 From SAM's Fixed Radius to a Learned Geometry

Across every structure treated so far, $s = +1$ behaves uniformly well; $s = -1$ is where the trouble sits, its caps pulling in opposite directions (Table 8) and its LRD/BD safeguard only conditional (§6.3), unlike the unconditional guarantee for $s = +1$. We fix $s = -1$ for the rest of this subsection, since the lower CBO underlies Sharpness-Aware Minimization (SAM) [31], which replaces the training loss with a worst-case loss over a ball of radius ρ around θ. With SAM's weight decay read as an L^2 prior of precision δ,

$$\mathcal{E}_{\mathrm{SAM}}(\theta;\rho,\delta) \coloneqq \sup_{\|\varepsilon\|_2\le\rho} \ell(\theta+\varepsilon) + \tfrac{\delta}{2}\|\theta\|_2^2. \tag{18}$$

The indicator candidate is exactly SAM. The structural connection between the lower CBO and SAM established in Propositions 7.1 and 7.2 was first derived by Shelvia Wongso [9].

Proposition 7.1 (SAM as a lower CBO under an indicator candidate [9])**.** *Let $\Theta \subseteq \mathbb{R}^d$ be closed, ℓ continuous, and $\pi \equiv \mathbf{1}_\Theta$ the uninformative prior. Take $g_{\eta,\rho}(\theta) \coloneqq \mathbf{1}[\|\theta-\eta\|_2 \le \rho]$, where $\mathbf{1}[\cdot]$ is the indicator function of an event, and let $\bar{B}(\eta,\rho) \coloneqq \{\theta \in \mathbb{R}^d : \|\theta-\eta\|_2 \le \rho\}$ denote the closed Euclidean ball with $\eta \in \Theta$ and $\rho > 0$ fixed. Then $g_{\eta,\rho} \in \mathcal{F}(\Theta)$ and, with $\log 0 = -\infty$,*

$$\underline{\mathrm{CBO}}(g_{\eta,\rho}) = -\sup_{\theta\in\Theta\cap\bar{B}(\eta,\rho)} \ell(\theta),$$

attained, so

$$\arg\max_{\eta} \underline{\mathrm{CBO}}(g_{\eta,\rho}) = \arg\min_{\eta} \mathcal{E}_{\mathrm{SAM}}(\eta;\rho,0), \tag{19}$$

an identity of objective functions, not just of stationary points.

If moreover $\nabla\ell(\eta) \neq 0$ and $B(\eta,\rho) \subseteq \Theta$, the first-order local model $\ell(\eta+\varepsilon) \approx \ell(\eta) + \nabla\ell(\eta)^\top\varepsilon$ has unique maximiser over $\|\varepsilon\|_2 \le \rho$

$$\varepsilon^\star_{\mathrm{SAM}} = \rho\,\frac{\nabla\ell(\eta)}{\|\nabla\ell(\eta)\|_2}, \tag{20}$$

exact when ℓ is affine on $B(\eta,\rho)$, else the first-order approximation to the true maximiser.

Proof. $g_{\eta,\rho}(\eta) = 1$ gives $\sup_\theta g_{\eta,\rho} = 1$, so $g_{\eta,\rho} \in \mathcal{F}(\Theta)$. Off $\Theta \cap \bar{B}(\eta,\rho)$, $\log g_{\eta,\rho} = -\infty$ so the inner term is $+\infty$, excluded from the infimum since it is finite on $\Theta \cap \bar{B}(\eta,\rho)$ (nonempty, contains η). On $\Theta\cap\bar{B}(\eta,\rho)$ the term is $-\ell(\theta)$, giving $\underline{\mathrm{CBO}}(g_{\eta,\rho}) = -\sup_{\Theta\cap\bar{B}(\eta,\rho)} \ell$, attained since $\Theta\cap\bar{B}(\eta,\rho)$ is compact and ℓ continuous. Substituting $\theta = \eta+\varepsilon$ and negating gives (19).

For (20): under $B(\eta,\rho) \subseteq \Theta$ the constraint reduces to $\|\varepsilon\|_2 \le \rho$, and the linear model $\ell(\eta)+\nabla\ell(\eta)^\top\varepsilon$ is maximised, by Cauchy–Schwarz, uniquely at ε a nonnegative multiple of $\nabla\ell(\eta)$ with $\|\varepsilon\|_2 = \rho$, giving $\varepsilon^\star_{\mathrm{SAM}} = \rho\,\nabla\ell(\eta)/\|\nabla\ell(\eta)\|_2$. □

From a fixed radius to a learned geometry. Proposition 7.1 draws an exact analogy between the lower CBO of the indicator candidate and SAM, but its radius ρ is fixed by hand. Following [9], consider instead $g_{\eta,\vartheta}(\theta) = \mathcal{N}^*(\theta;\eta,\vartheta^{-1}I) = \exp(-\frac{\vartheta}{2}\|\theta-\eta\|_2^2)$, whose precision ϑ Sections 6.1 and 6.2 learn. This switch gives up the exact identity, since the Gaussian candidate decays smoothly with no fixed boundary, but its superlevel set at any level $\alpha \in (0,1)$ is itself a closed Euclidean ball [9]:

$$\{\theta \in \mathbb{R}^d : g_{\eta,\vartheta}(\theta) \ge \alpha\} = \bar{B}(\eta,\rho_\alpha), \qquad \rho_\alpha \coloneqq \vartheta^{-1/2}\sqrt{-2\log\alpha}. \tag{21}$$

Fixing $\alpha = e^{-1/2}$ from now on gives $\rho_\alpha = \vartheta^{-1/2}$. Rather than learning a radius directly, we instead seek to learn ϑ, which sets the radius at this fixed level.

Proposition 7.2 (Gaussian lCBO and its perturbation [9]). *Let $\Theta = \mathbb{R}^d$ and $\pi \equiv \mathbf{1}_\Theta$. Then*

$$\underline{\mathrm{CBO}}(g_{\eta,\vartheta}) = -\sup_{\theta\in\mathbb{R}^d}\big\{\ell(\theta) - \tfrac{\vartheta}{2}\|\theta-\eta\|_2^2\big\},$$

the analogue of $\underline{\mathrm{CBO}}(g_{\eta,\rho}) = -\sup_{\|\theta-\eta\|_2\le\rho}\ell(\theta)$ with the hard constraint relaxed to a quadratic penalty. If $\nabla\ell$ is globally L-Lipschitz with $L<\vartheta$, the maximiser $u^\star$ of $u\mapsto\ell(\eta+u)-\frac{\vartheta}{2}\|u\|_2^2$ exists and is unique. Under the first-order local model $\ell(\eta+u)\approx\ell(\eta)+\nabla\ell(\eta)^\top u$, the maximiser of the resulting surrogate is

$$u^\star_{\mathrm{soft}} = \vartheta^{-1}\nabla\ell(\eta), \qquad \|u^\star_{\mathrm{soft}}\|_2 = \vartheta^{-1}\|\nabla\ell(\eta)\|_2, \tag{22}$$

equal to $u^\star$ when ℓ is affine near η, else its first-order approximation.

Proof. Substituting $-\log g_{\eta,\vartheta} = \frac{\vartheta}{2}\|\theta-\eta\|_2^2$ into $\underline{\mathrm{CBO}}$ gives $-\sup_\theta\{\ell(\theta)-\frac{\vartheta}{2}\|\theta-\eta\|_2^2\}$. Set $\theta=\eta+u$, $\phi(u) \coloneqq \ell(\eta+u)-\frac{\vartheta}{2}\|u\|_2^2$. The descent lemma from L-Lipschitz $\nabla\ell$ gives ϕ $(\vartheta-L)$-strongly concave for $\vartheta>L$, hence coercive with a unique global maximiser $u^\star$. Under the first-order model, the objective becomes $\ell(\eta)+\nabla\ell(\eta)^\top u-\frac{\vartheta}{2}\|u\|_2^2$, whose gradient vanishes uniquely at $u^\star_{\mathrm{soft}}=\vartheta^{-1}\nabla\ell(\eta)$, with Hessian $-\vartheta I\prec 0$. This coincides with $u^\star$ when ℓ is affine near η, since the approximation is then exact. □

As noted in [9], $\varepsilon^\star_{\mathrm{SAM}}$ and $u^\star_{\mathrm{soft}}$ share the direction $\nabla\ell(\eta)$ but not the norm: ρ is fixed regardless of the local gradient, while $\|u^\star_{\mathrm{soft}}\|_2=\vartheta^{-1}\|\nabla\ell(\eta)\|_2$ grows with the gradient and shrinks with ϑ.

The induced radius. Using the bridge of (21) at $\alpha=e^{-1/2}$, and with the log-volume penalty in place, Prop. 6.10 gives $\vartheta^\star = d/\|u_{-1}\|_2^2$ with Newton step $u_{-1}\approx -H_{-1}^{-1}\nabla\hat\ell(\eta)$, so the induced radius $\rho_\alpha=\vartheta^{-1/2}$ is

$$\rho^\star(\eta) \approx \frac{1}{\sqrt{d}}\,\big\|H_{-1}^{-1}\nabla\hat\ell(\eta)\big\|_2. \tag{23}$$

Through this relaxation, we attempt to learn SAM's own radius ρ by reusing the candidate whose precision Section 6 already learns, trading the tuning of ρ, whose optimal magnitude scales with unknown curvature, for the safeguard parameters r and ϵ_{rel} of §6.2. No separate optimiser is needed, as the isotropic update at $s=-1$ already produces $\vartheta^\star$, from which $\rho^\star(\eta)$ is immediate. Whether this relaxation is theoretically sound is left open. We test what it yields experimentally instead (§7.3).

7.2 Practical Algorithms and Computational Complexity

We now assemble the isotropic and diagonal stationarity conditions of Proposition 6.10, together with the signed caps $\mathcal{C}_{\pm1}$ of (14) and the damped Picard steps of (13) (Section 6.2), into the two practical optimisers below.

Algorithm 1 `CBOpt-iso`

Require: dimension d, $s\in\{\pm1\}$, $\{\rho_t\}$, $\delta>0$, $r,\epsilon_{\mathrm{rel}}>0$, $\beta_1,\beta_2\in[0,1)$, $h_0>0$, $w\in(0,1]$, inner steps $K\ge1$

$\eta\leftarrow$ weights, $h\leftarrow h_0\mathbf{1}$, $g\leftarrow\mathbf{0}$, $\vartheta\leftarrow(h_0+\delta)(1-s\,\epsilon_{\mathrm{rel}})$
for $t=1,2,\dots$ **do**
  $\hat g\leftarrow\widehat{\nabla\ell}(\eta)$, $g\leftarrow\beta_1 g+(1-\beta_1)\hat g$, $h\leftarrow\beta_2 h+(1-\beta_2)\hat g^{\odot2}$, $\bar g\leftarrow g/(1-\beta_1^t)$
  $m\leftarrow\min_j(h_j+\delta)$, $M\leftarrow\max_j(h_j+\delta)$
  for $k=1,\dots,K$ **do**
    $u\leftarrow -s(\mathrm{diag}(h+\delta)-\vartheta I)^{-1}(\bar g+\delta\eta)$
    $\vartheta^\star\leftarrow d/\|u\|^2$ ▷ Prop. 6.10
    if $s=+1$ **then**
      $\vartheta^\star_{\mathrm{c}}\leftarrow\max\big(\min(\vartheta^\star,\ m/r,\ m(1-\epsilon_{\mathrm{rel}})),\ \epsilon_{\mathrm{rel}}\,m\big)$ ▷ $\mathcal{C}_{+1}$, (14) with $v_{\min}=\epsilon_{\mathrm{rel}}m$
    else
      $\vartheta^\star_{\mathrm{c}}\leftarrow\max\big(\min(\vartheta^\star,\ M/r),\ M(1+\epsilon_{\mathrm{rel}})\big)$ ▷ $\mathcal{C}_{-1}$, (14)
    end if
    $\vartheta\leftarrow(1-w)\vartheta+w\vartheta^\star_{\mathrm{c}}$ ▷ (13)
  end for
  $\eta\leftarrow\eta+\rho_t u$
end for
return η,ϑ

Algorithm 2 `CBOpt-diag`

Require: dimension d, $s \in \{\pm 1\}$, $\{\rho_t\}$, $\delta > 0$, $r, \epsilon_{\mathrm{rel}} > 0$, $\beta_1, \beta_2 \in [0,1)$, $h_0 > 0$, $w \in (0,1]$, inner steps $K \geq 1$

$\eta \leftarrow$ weights, $h \leftarrow h_0\mathbf{1}$, $g \leftarrow \mathbf{0}$, $\boldsymbol{\vartheta} \leftarrow (h_0+\delta)(1 - s\,\epsilon_{\mathrm{rel}})\mathbf{1}$
for $t = 1, 2, \ldots$ **do**
  $\hat{g} \leftarrow \widehat{\nabla}\ell(\eta)$, $g \leftarrow \beta_1 g + (1-\beta_1)\hat{g}$, $h \leftarrow \beta_2 h + (1-\beta_2)\hat{g}^{\odot 2}$, $\bar{g} \leftarrow g/(1-\beta_1^t)$
  for $k = 1, \ldots, K$ **do**
    $u \leftarrow -s\,(h + \delta - \boldsymbol{\vartheta})^{-1} \odot (\bar{g} + \delta\eta)$
    $\boldsymbol{\vartheta}^\star \leftarrow 1/(u \odot u)$ ▷ Prop. 6.10
    if $s = +1$ **then**
      $\boldsymbol{\vartheta}_{\mathrm{c}}^\star \leftarrow \max\big(\min(\boldsymbol{\vartheta}^\star,\ (h+\delta)/r,\ (h+\delta)(1-\epsilon_{\mathrm{rel}})),\ \epsilon_{\mathrm{rel}}\,(h+\delta)\big)$ ▷ $\mathcal{C}_{+1}^{\mathrm{diag}}$, (14)
    else
      $\boldsymbol{\vartheta}_{\mathrm{c}}^\star \leftarrow \max\big(\min(\boldsymbol{\vartheta}^\star,\ (h+\delta)/r),\ (h+\delta)(1+\epsilon_{\mathrm{rel}})\big)$ ▷ $\mathcal{C}_{-1}^{\mathrm{diag}}$, (14)
    end if
    $\boldsymbol{\vartheta} \leftarrow (1-w)\boldsymbol{\vartheta} + w\boldsymbol{\vartheta}_{\mathrm{c}}^\star$ ▷ (13)
  end for
  $\eta \leftarrow \eta + \rho_t\, u$
end for
return $\eta, \boldsymbol{\vartheta}$

Both algorithms share an EMA Newton step u, a precision target, a cap enforcing $H_s \succ 0$, and K damped Picard steps (Theorem 6.14), with `CBOpt-diag` acting coordinatewise. Setting $v_{\min} = \epsilon_{\mathrm{rel}} m$ (resp. $\epsilon_{\mathrm{rel}}(h_j + \delta)$) scales the floor to local curvature, ensuring $\mathcal{I}_{+1} \neq \emptyset$ for $\epsilon_{\mathrm{rel}} \in (0, 1/2)$, while $\mathcal{I}_{-1} \neq \emptyset$ requires $r < 1/(1+\epsilon_{\mathrm{rel}})$.

For the structured precisions of Section 6.3, the rank-one target (Proposition 6.18) is regularised to the LRD form (16). Positivity of H_s is unconditional for $s = +1$ (Theorem 6.20) but conditional for $s = -1$ (Theorem 6.21), requiring the feasibility test and sign flip of Remark 6.22. `CBOpt-LRD` uses a global precision, while `CBOpt-BD` acts block-wise (Theorem 6.24).

When the $s = -1$ feasibility test fails, no $\psi > 0$ can certify $H_s \succ 0$ via Theorem 6.21. We therefore flip to $s_{\mathrm{e}} = +1$ instead of skipping the step. Following the standard warm-up principle of starting broad before tightening toward the target [32], the support-covering upper CBO keeps updating η when u is large, restoring the mode-seeking lower CBO once $\|u\|^2$ becomes feasible. For `CBOpt-BD`, this flip applies per block, allowing a mixed-sign objective.

Algorithm 3 `CBOpt-LRD`

Require: $s \in \{\pm 1\}$, $\{\rho_t\}$, $\delta > 0$, $a, r > 0$, $\xi \in (0,1)$, $\varphi > 1$, $\beta_1, \beta_2 \in [0,1)$, $h_0 > 0$

$\eta \leftarrow$ weights, $h \leftarrow h_0\mathbf{1}$, $g \leftarrow \mathbf{0}$, $S \leftarrow \delta I$, $s_{\mathrm{e}} \leftarrow s$
for $t = 1, 2, \ldots$ **do**
  $\hat{g} \leftarrow \widehat{\nabla}\ell(\eta)$, $g \leftarrow \beta_1 g + (1-\beta_1)\hat{g}$, $h \leftarrow \beta_2 h + (1-\beta_2)\hat{g}^{\odot 2}$, $\bar{g} \leftarrow g/(1-\beta_1^t)$
  $m \leftarrow \min_j(h_j + \delta)$, $M \leftarrow \max_j(h_j + \delta)$
  $u_{\max}^2 \leftarrow (1-\xi)/M$ ▷ Rem. 6.22, (17)
  $u \leftarrow -s_{\mathrm{e}}(\mathrm{diag}(h+\delta) - S)^{-1}(\bar{g} + \delta\eta)$
  if $s = -1 \ \wedge\ \|u\|^2 \leq u_{\max}^2$ **then**
    $s_{\mathrm{e}} \leftarrow -1$
  else if $s = -1$ **then**
    $s_{\mathrm{e}} \leftarrow +1$ ▷ Rem. 6.22
  end if
  if $s_{\mathrm{e}} = +1$ **then**
    $\psi \leftarrow \max(r,\ 1/m + a)$ ▷ Thm. 6.20
  else
    $\psi \leftarrow \max(r,\ \varphi^{-1}(1/M - \|u\|^2))$ ▷ Thm. 6.21
  end if
  $S \leftarrow (uu^\top + \psi I)^{-1}$, $\eta \leftarrow \eta + \rho_t u$
end for
return η, S

Algorithm 4 `CBOpt-BD`

Require: blocks $\ell \in \{1, \dots, L\}$, $s \in \{\pm 1\}$, $\{\rho_t\}$, $\delta > 0$, $a, r > 0$, $\xi \in (0,1)$, $\varphi > 1$, $\beta_1, \beta_2 \in [0,1)$, $h_0 > 0$

$\eta \leftarrow$ weights, $h \leftarrow h_0 \mathbf{1}$, $g \leftarrow \mathbf{0}$, $S_\ell \leftarrow \delta I_{d_\ell}$, $s_{\mathrm{e},\ell} \leftarrow s$
for $t = 1, 2, \dots$ **do**
  $\hat{g} \leftarrow \widehat{\nabla \ell}(\eta)$, $g \leftarrow \beta_1 g + (1-\beta_1)\hat{g}$, $h \leftarrow \beta_2 h + (1-\beta_2)\hat{g}^{\odot 2}$, $\bar{g} \leftarrow g/(1-\beta_1^t)$
  for $\ell = 1, \dots, L$ **do**
    $m_\ell, M_\ell \leftarrow \min_{j \in \ell}(h_j + \delta),\ \max_{j \in \ell}(h_j + \delta),\ u^2_{\max,\ell} \leftarrow (1-\xi)/M_\ell$ ▷ Rem. 6.22
    $u_\ell \leftarrow -s_{\mathrm{e},\ell}(\mathrm{diag}(h_\ell + \delta) - S_\ell)^{-1}(\bar{g}_\ell + \delta \eta_\ell)$
    if $s = -1 \ \wedge\ \|u_\ell\|^2 \le u^2_{\max,\ell}$ **then**
      $s_{\mathrm{e},\ell} \leftarrow -1$
    else if $s = -1$ **then**
      $s_{\mathrm{e},\ell} \leftarrow +1$ ▷ Rem. 6.22
    end if
    if $s_{\mathrm{e},\ell} = +1$ **then**
      $\psi_\ell \leftarrow \max(r,\ 1/m_\ell + a)$ ▷ Thm. 6.24(a)
    else
      $\psi_\ell \leftarrow \max(r,\ \varphi^{-1}(1/M_\ell - \|u_\ell\|^2))$ ▷ Thm. 6.24(b)
    end if
    $S_\ell \leftarrow (u_\ell u_\ell^\top + \psi_\ell I)^{-1}$
  end for
  $\eta \leftarrow \eta + \rho_t(u_1, \dots, u_L)$
end for
return $\eta, \{S_\ell\}$

Eight named optimisers follow: `uCBOpt-iso`, `lCBOpt-iso`, `uCBOpt-diag`, `lCBOpt-diag`, `uCBOpt-LRD`, `lCBOpt-LRD`, `uCBOpt-BD`, and `lCBOpt-BD`, with `u`/`l` denoting $s = \pm 1$. Unlike ϑ_0 in §6.2, the initial $S \leftarrow \delta I$ requires no careful choice: it merely ensures a valid $S \succ 0$ before computing ψ in closed form.

Complexity The outer mode recursion in η is left unanalysed (Remark 6.17), run for a fixed epoch budget rather than a convergence criterion, as in [1]. Within a single mode step, `CBOpt-iso` and `CBOpt-diag` take K damped Picard steps to solve the implicit fixed point for ϑ, while `CBOpt-LRD` and `CBOpt-BD` compute ψ (resp. ψ_ℓ) in closed form. The stochastic gradient and EMA update g_t, h_t cost the same forward and backward pass [33] for all four. What distinguishes them is how S is stored and how $H_s^{-1}(\bar{g}_t + \delta\eta_t)$ is computed, $O(d^2)$ and $O(d^3)$ for a naive dense $S \in \mathcal{S}^d_{++}$ recomputed at every step.

Proposition 7.3 (Storage and inversion complexity, single step)**.** *Per mode step, storing the precision costs $O(1)$ for `CBOpt-iso` and $O(d)$ for `CBOpt-diag`, `CBOpt-LRD`, and `CBOpt-BD` ($O(d_\ell)$ per block, summed across blocks: $\sum_\ell O(d_\ell) = O(d)$ overall). Computing $H_s^{-1}(\bar{g}_t + \delta\eta_t)$ once costs $O(d)$ for all four candidates.*

Proof. Storage is immediate: a scalar ϑ ($O(1)$) for the isotropic candidate, $\boldsymbol{\vartheta} \in \mathbb{R}^d$ ($O(d)$) for the diagonal candidate, and, by Lemma 6.19, $S^{-1} = uu^\top + \psi I$ stored as $(u, \psi) \in \mathbb{R}^d \times \mathbb{R}_{>0}$ ($O(d)$) for LRD, summed over the L blocks for BD ($\sum_\ell O(d_\ell) = O(d)$). For inversion, $S = \frac{1}{\psi} I - b\, uu^\top$ with $b = 1/(\psi(\psi + \|u\|^2))$. Writing $P = \mathrm{diag}(h + \delta) - \frac{1}{\psi} I$, strictly invertible since $1/\psi < h_j + \delta$ ($s = +1$) and $1/\psi > h_j + \delta$ ($s = -1$), $H_s = s(P + b\, uu^\top)$. Sherman–Morrison gives, with $z := \bar{g}_t + \delta\eta_t$, $H_s^{-1} z = s\big(P^{-1}z - b(P^{-1}u)(u^\top P^{-1}z)/(1 + b\, u^\top P^{-1}u)\big)$. The denominator never vanishes: Theorems 6.20 and 6.21 guarantee $H_s \succ 0$, so $sH_s = P + b\, uu^\top$ is invertible, and by the matrix determinant lemma $\det(P)(1 + b\, u^\top P^{-1}u) = \det(P + b\, uu^\top) \neq 0$; since $\det(P) \neq 0$, this forces $1 + b\, u^\top P^{-1}u \neq 0$. Diagonal P makes every operation coordinatewise, costing $O(d)$ with no dense $d \times d$ matrix formed. BD applies the same solve per block ($O(d)$ overall), while isotropic and diagonal inversions are directly $O(d)$, H_s being diagonal in both cases. □

Computationally, `CBOpt-LRD` and `CBOpt-BD` are the cheapest per mode step: no iteration on ϑ (resp. S_ℓ), just the single $O(d)$ solve above. `CBOpt-iso` and `CBOpt-diag` instead take K such solves per mode step, so their per-mode-step cost is $O(Kd)$, against $O(d)$ for `CBOpt-LRD` and `CBOpt-BD`, predicting the ordering LRD, BD < iso, diag.

7.3 Experimental validation on an exact toy problem and LeNet-5/Fashion-MNIST

The constructions of §6.1–7.1 are tested in two settings. First, a deterministic isotropic quadratic toy problem isolates the penalty mechanism, stationarity condition, and SAM direction from curvature-estimation noise, as all quantities in Assumption 6.1 and Propositions 6.10 and 6.7 are exact. Second, training LeNet-5 on Fashion-MNIST against EMNIST-Letters (Section 4) evaluates benchmark performance, curvature safeguards, the Picard contraction constant ($K = 5$ damped steps), and the learned SAM radius.

Hyperparameters, diagnostics, implementation details, and corresponding **repository link** are given in Appendix A. The benchmark uses three seeds (matching the reproduction protocol), while remaining experiments use a single seed to report exact or per-step certified metrics. Curvature $H_s \succ 0$ is monitored strictly before capping and safeguards, since Theorems 6.20–6.24 guarantee it thereafter.

The toy problem. The loss is a diagonal quadratic $\ell(\theta) = \frac{1}{2}(\theta - \theta_0)^\top \mathrm{diag}(\boldsymbol{c})(\theta - \theta_0)$ in $d = 30$, with log-uniformly sampled $\boldsymbol{c}$ and prior precision $\delta = 2 \times 10^{-3}$. Two regimes are tested via Gaussian perturbations η of θ_0 (near and far). Here $\theta_s(\vartheta)$, $u_s(\vartheta)$, CBO_s, the penalised objective, and $\vartheta^\star$ are all available in closed form without minibatch or EMA noise. A separate dimension sweep ($d \in [3.5 \times 10^3, 1.9 \times 10^5]$, averaged over six runs) evaluates the complexity scaling of Proposition 7.3.

Toy results. Five results are reported on the toy problem, summarised below:

- **Regularity (iso, $s = \pm 1$).** On the isotropic candidate, $\theta_s(\vartheta)$ exists and stays finite (hence continuous, by Lemma 6.6) across a 400-point grid of ϑ, with $u_s(\vartheta) \neq 0$ throughout, confirming Assumption 6.1(d) numerically in both the far and near regimes and for both signs. Uniqueness follows analytically from the strict concavity/convexity of the inner objective whenever ℓ is quadratic (Proposition 6.10's proof).
- **Monotonicity and its repair (iso, $s = \pm 1$).** The raw CBO_s is strictly monotone in ϑ in every case, confirming Lemma 6.3. With the log-volume penalty added, the objective instead acquires an interior stationary point in every case (Proposition 6.7, Theorem 6.5), with $\vartheta^\star = 0.16$ and 18.2 for $s = \pm 1$ in the far regime, 0.23 and 4.5 in the near regime (see Figure 4).

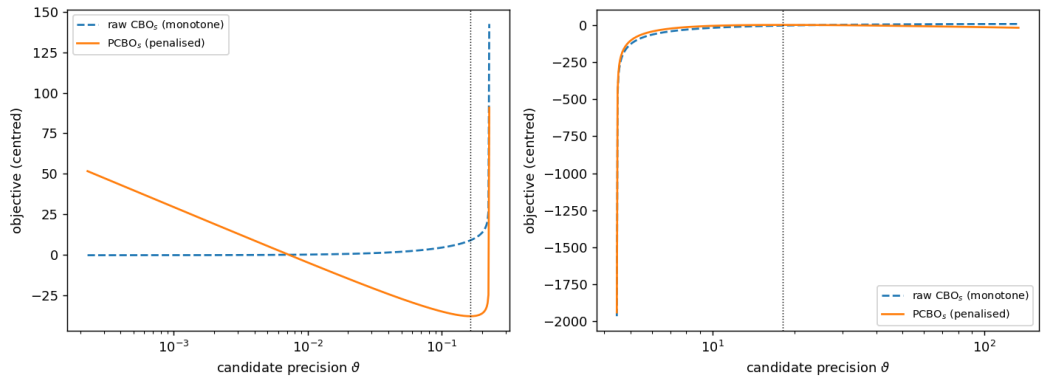


Figure 4: Toy problem, isotropic quadratic ($d = 30$), far regime. The log-volume penalty breaks the strict monotonicity of the raw CBO_s, inducing a clear interior minimum for $s = +1$ and a modest maximum for $s = -1$.

The $s = -1$ maximum is barely visible on this scale, since the plot spans a wide enough range to show PCBO_s diverging toward $-\infty$ on both sides, on the left as $\vartheta \to \lambda_{\max}(\nabla^2 \hat{\ell})^+$ (driven by the inner infimum) and on the right as $\vartheta \to \infty$ (driven by the log-volume penalty). Since ℓ is exactly quadratic here, Proposition 6.7 guarantees this maximum, like the $s = +1$ minimum, is global rather than merely stationary.

- **Sensitivity to the penalty coefficient (iso, $s = \pm 1$).** Scaling the coefficient over $(\frac{1}{4}, \frac{1}{2}, 1, 2, 4)$ shifts $\vartheta^\star$ monotonically from 0.074 to 0.21 for $s = +1$ (2.8×) and from 55 to 8.5 for $s = -1$ (6.4×). The marked sensitivity of $s = -1$ confirms that $\vartheta^\star$ is not robust to arbitrary scaling, justifying fixing $\zeta = d/2$ via the volume argument (Proposition 6.9).

- **SAM direction (iso, $s = -1$, far regime).** Reusing $\vartheta^\star = 18.2$, a sweep over $\rho \in \{0.05, 0.2, 1\}$ confirms that $\varepsilon^\star_{\mathrm{SAM}}$ and $u^\star_{\mathrm{soft}}$ remain collinear to machine precision (cos = 1.000000, Propositions 7.1, 7.2). Only the norms diverge: $\|\varepsilon^\star_{\mathrm{SAM}}\| = \rho$ tracks the swept radius, while $\|u^\star_{\mathrm{soft}}\| = 1.05$ and $\rho_\alpha = 0.23$ stay fixed.

- **Dimension sweep (LRD, BD, iso, diag).** Per-step runtime (Table 9, 6 runs) increases from a few milliseconds at $d \approx 3 \times 10^3$ to 5–12 ms at $d \approx 2 \times 10^5$. The ranking LRD < BD < iso < diag holds at four of the five dimensions, matching the predicted $O(d)$ versus $O(Kd)$ complexity ($K = 5$ Picard steps). The gap is closer to 2× than the nominal 5×, reflecting GPU kernel-launch overheads alongside numerical cost at this scale.

Table 9: Toy dimension sweep, mean ms/step over 6 runs (± std). Bold entries indicate deviations from the usual ordering LRD < BD < iso < diag.

d	LRD	BD	iso	diag
3,466	1.47 ± 0.04	2.30 ± 0.12	3.27 ± 0.17	3.86 ± 0.32
8,970	1.63 ± 0.18	2.33 ± 0.15	3.24 ± 0.11	4.20 ± 0.31
26,122	2.40 ± 0.11	3.63 ± 0.21	3.88 ± 0.34	7.13 ± 0.27
85,002	2.75 ± 0.20	3.50 ± 0.06	$\mathbf{8.07 \pm 0.55}$	$\mathbf{7.96 \pm 2.01}$
190,410	5.05 ± 0.50	5.28 ± 0.24	10.05 ± 0.71	12.01 ± 0.42

Remark 7.4 (What the precision structures represent). Extending Remark 4.1: the isotropic scalar ϑ assigns uniform confidence across all weights. The diagonal precision $\mathrm{diag}(\boldsymbol{\vartheta})$ allows confidence to vary coordinatewise along parameter axes. A full precision $S \in \mathcal{S}^d_{++}$ captures directional interactions (such as high certainty along $w_1 + w_2$ but low along $w_1 - w_2$), while layer-wise block-diagonal candidates capture these interactions within each layer independently.

The real setting. LeNet-5 on Fashion-MNIST (in-domain) against EMNIST-Letters (OOD) follows the protocol of Section 4 (§4.1). Four analyses evaluate the optimisers of §7.2, with remaining hyperparameters detailed in Appendix A:

- **Benchmark.** `uCBOpt-iso`, `uCBOpt-diag`, and `uCBOpt-BD` ($s = +1$, learned precision, three seeds, 100 epochs). Theorem 6.20 makes $s = +1$ the safe default; `uCBOpt-LRD` is omitted as the single-block instance of BD. Results are compared directly against the baseline numbers (fixed `uCBOpt`, AdamW, IVON) reused from Section 4.

- **Geometry check.** `lCBOpt-LRD` and `lCBOpt-BD` ($s = -1$, one seed, 50 epochs, diagnostics only) test whether the $s = -1$ safeguard interval is ever feasible at this scale. The $s = +1$ rates require no dedicated run, being logged directly during the benchmark group's training.

- **SAM application.** Six runs (one seed, 50 epochs each): three `lCBOpt-iso` runs sweeping the anti-collapse tolerance $r \in \{1, 0.95, 1.05\} \times r_{\mathrm{nom}}$ ($r_{\mathrm{nom}} = 0.50$), and three standard SAM runs sweeping $\rho \in \{1, 0.95, 1.05\} \times \bar{\rho}_{\mathrm{nom}}$, where $\bar{\rho}_{\mathrm{nom}}$ is the temporal mean of $\vartheta_t^{-1/2}$ along the nominal run (within 5% of its final value), evaluating whether the learned radius avoids manual tuning.

- **Picard contraction constant.** Evaluated without retraining: $|\Phi'|$ and its explicit bound L_Φ (Remark 6.15) are measured along the trajectories already logged by the benchmark ($s = +1$) and SAM ($s = -1$) runs above.

Benchmark results. Table 10 reports the final evaluation metrics and Figure 5 displays the training curves across the benchmark group.

Table 10: Learned-precision benchmark, mean ± std over three seeds (`uCBOpt-BD` aggregates the two seeds that did not diverge). Bold marks the best value per metric within each block (learned-precision variants above, baselines below).

	Acc ↑	NLL ↓	ECE ↓	FPR@95 ↓	AUROC ↑
`uCBOpt-iso` ($s = +1$, learned ϑ)	0.912 ± 0.001	0.257 ± 0.002	0.022 ± 0.001	$\mathbf{0.263 \pm 0.015}$	$\mathbf{0.793 \pm 0.018}$
`uCBOpt-diag` ($s = +1$, learned $\boldsymbol{\vartheta}$)	$\mathbf{0.915 \pm 0.001}$	$\mathbf{0.252 \pm 0.001}$	0.020 ± 0.002	0.292 ± 0.018	0.758 ± 0.018
`uCBOpt-BD` ($s = +1$, learned S)	0.908 ± 0.002	0.256 ± 0.001	$\mathbf{0.014 \pm 0.001}$	0.289 ± 0.017	0.755 ± 0.018
`uCBOpt` (fixed ϑ)	0.909 ± 0.002	0.272 ± 0.004	0.025 ± 0.001	0.263 ± 0.005	0.795 ± 0.005
IVON	$\mathbf{0.910 \pm 0.002}$	$\mathbf{0.258 \pm 0.003}$	$\mathbf{0.019 \pm 0.001}$	$\mathbf{0.241 \pm 0.009}$	$\mathbf{0.812 \pm 0.012}$
AdamW	0.896 ± 0.003	0.296 ± 0.005	0.022 ± 0.006	0.335 ± 0.005	0.722 ± 0.016

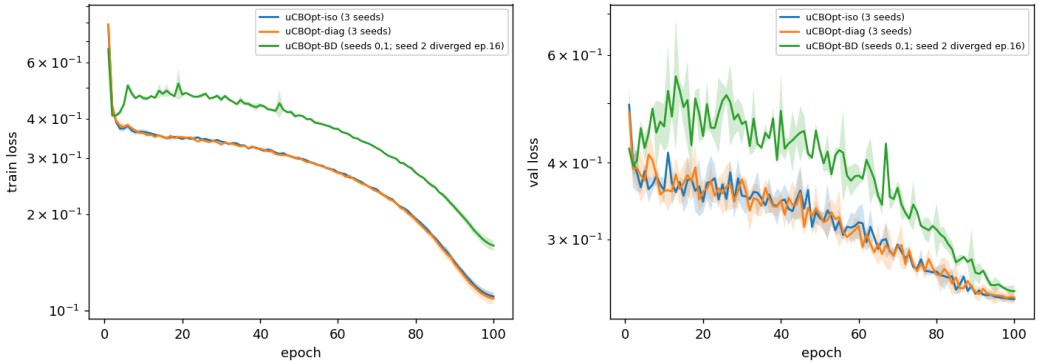


Figure 5: Training and validation loss, mean ± std over seeds. LeNet-5 / Fashion-MNIST, 100 epochs.

As in Section 4, AdamW has no candidate distribution, IVON learns both mode and a diagonal variance proxy, and fixed-ϑ `uCBOpt` [1] serves as the direct ablation. Against it, `uCBOpt-iso` and `uCBOpt-diag` improve accuracy (0.912 ± 0.001 and 0.915 ± 0.001 against 0.909 ± 0.002), while `uCBOpt-BD` is indistinguishable (0.908 ± 0.002). All three learned variants improve NLL (down to 0.252–0.257 against 0.272 ± 0.004) and lower ECE (0.014–0.022 against 0.025 ± 0.001), since adapting ϑ to local curvature calibrates confidence far better than the fixed $\vartheta = 8 \times 10^{-6}$ baseline, even though ϑ operates at the cap boundary rather than an interior optimum (**Geometry and safeguards**). OOD detection is the exception: unchanged for `uCBOpt-iso`, but noticeably degraded for `uCBOpt-diag` and `uCBOpt-BD`.

Among the baselines, IVON performs best across all five metrics. Relative to the learned-precision models, however, its advantage is strictly confined to OOD detection (FPR@95 of 0.241 ± 0.009, AUROC of 0.812 ± 0.012). On in-domain metrics, `uCBOpt-diag` beats IVON on accuracy (0.915 ± 0.001 against 0.910 ± 0.002) with non-overlapping standard deviations, all three learned variants surpass IVON on NLL (0.258 ± 0.003), and `uCBOpt-BD` clearly outperforms it on ECE (0.014 ± 0.001 against 0.019 ± 0.001). Seed variance reveals a consistent pattern: all three learned-precision variants are tighter than fixed `uCBOpt` in-domain, but show wider dispersion on OOD metrics (0.015–0.018 against 0.005). Because IVON exhibits the exact same spread increase on OOD (0.001–0.003 widening to 0.009–0.012), this reflects standard model underspecification [34] under learned dispersion rather than an issue with possibilistic inference itself.

One of three `uCBOpt-BD` seeds diverged at epoch 16 from a near-singular per-block solve (Appendix A), making it the most numerically fragile variant. `uCBOpt-BD` ends with higher training loss than `uCBOpt-iso`/`uCBOpt-diag` (Figure 5), a gap that persists throughout training and points to a structural constraint of block-diagonalisation rather than unfinished optimisation. However, `uCBOpt-BD` closely matches them on validation loss, so this training gap does not hurt generalisation. In summary, performance trades off across structures: `uCBOpt-diag` is strongest on accuracy and NLL, `uCBOpt-BD` provides the clear win on calibration (ECE), and `uCBOpt-iso` provides the most balanced overall profile.

Geometry and safeguards. Following Shelvia Wongso's suggestion to examine the caps' impact, two questions are addressed here: how often $H_s \succ 0$ held for the isotropic and diagonal benchmark runs ($s = +1$) before the cap of §6.2, and how often the $s = -1$ LRD/BD safeguards of §6.3 had to flip sign (see Appendix A).

$H_s \succ 0$ held on 0.0% of steps before capping for both `uCBOpt-iso` and `uCBOpt-diag` at the raw Picard fixed point (Proposition 6.10); the cap $\mathcal{C}_{+1}$ (Eq. (14)) alone guarantees this property. While ϑ sits at the boundary $(h_j + \delta)(1 - \epsilon_{\text{rel}})$ rather than the interior optimum derived in Proposition 6.10, the cap still tracks local curvature h_j across training, unlike the fixed $\vartheta = 8 \times 10^{-6}$ baseline, explaining the benchmark's gains.

Capped ϑ reduces to $(1 - \epsilon_{\text{rel}})\tilde{h}_j$ (with $\tilde{h}_j := h_j + \delta$), matching the fixed-fraction-of-curvature form $\vartheta_t = \gamma c_t$ of isotropic `uCBOpt-adapt` [1] (§3.4), where $1 - \epsilon_{\text{rel}}$ acts as γ and $\tilde{h}_j$ as c_t. The two differ in origin and smoothing. Specifically, γc_t is an explicit running minimum of $\tilde{h}_t$ damped by β_3, whereas ours is the capped Picard fixed point. Numerically, they are close on Fashion-MNIST/LeNet (`uCBOpt-iso` vs `uCBOpt-adapt` [1]): Acc 0.912 vs 0.910, NLL 0.257 vs 0.256, ECE 0.022 vs 0.016, FPR@95 0.263 vs 0.272, AUROC 0.793 vs 0.773. However, while `uCBOpt-adapt` is restricted to scalar ϑ_t, this report develops and evaluates diagonal, LRD, and BD structures.

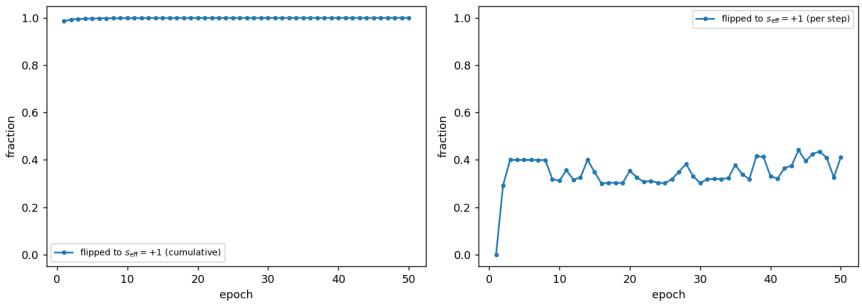
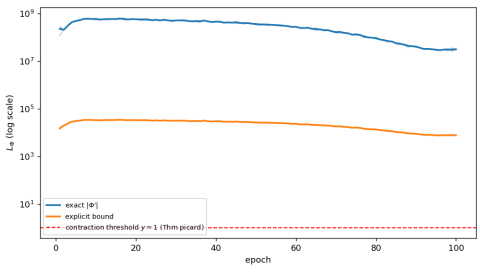


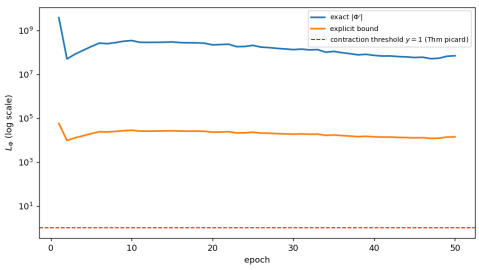
Figure 6: Flip rate to the effective sign $s_{\text{e}} = +1$, `lCBOpt-LRD` and `lCBOpt-BD` ($s = -1$), cumulative for `lCBOpt-LRD`, per-epoch for `lCBOpt-BD`.

`lCBOpt-LRD` flips on 100% of steps, $\|u\|^2 < 1/M$ across all d coordinates at once (Theorem 6.21) never satisfied at LeNet scale, so its $s = -1$ objective is never pursued. `lCBOpt-BD` flips far less, each block ℓ needing only $\|u_\ell\|^2 < 1/M_\ell$ (Theorem 6.24(b)): after rising from 0% at epoch 1 to about 0.40 by epoch 3, the flip fraction settles around 34.5%, so roughly 65.5% of blocks pursue the $s = -1$ objective at any step, the repair the per-block structure was designed to provide.

The Picard contraction constant. Figures 7a and 7b plot $|\Phi'|$ and its explicit estimate L_Φ of Remark 6.15 against epoch, for $s = +1$ (benchmark `uCBOpt-iso` run, three seeds) and $s = -1$ (first `lCBOpt-iso` SAM run, one seed).



(a) $s = +1$ (`uCBOpt-iso`, three seeds × 100 epochs), log scale, $y = 1$ the contraction threshold of Theorem 6.14.



(b) $s = -1$ (`lCBOpt-iso`, one seed × 50 epochs), same axes.

Both exceed 1 from the first epoch, reaching 3×10^7 ($s = +1$) and 7×10^7 ($s = -1$) by the end of training, $s = +1$ declining slightly but staying far above 1, $s = -1$ not declining at all. Theorem 6.14's sufficient condition fails by this margin at network scale, but it is evaluated at ϑ's current, capped value, not along a free Picard recursion: sitting at the cap's boundary rather than converging to an interior fixed point, ϑ never actually tests the regime the theorem certifies.

SAM application. The identity of Proposition 7.1, i.e. $\underline{\mathrm{CBO}}(g_{\eta,\rho}) = -\mathcal{E}_{\mathrm{SAM}}(\eta;\rho,0)$, is checked directly on a trained checkpoint. The gap between the projected-gradient-ascent estimate of $\sup_{\|\varepsilon\|_2\le\rho}\ell(\eta+\varepsilon)$ and the one-step $\ell(\eta+\varepsilon^\star_{\mathrm{SAM}})$ is 1.79×10^{-3} at $\rho=0.02$, 2.26×10^{-2} at $\rho=0.05$, and 1.12×10^{-1} at $\rho=0.1$, growing at an exponent close to 2, the $O(\rho^2)$ error expected since $\varepsilon^\star_{\mathrm{SAM}}$ is only a first-order approximation of the true maximiser.

Table 11 reports six runs, three possibilistic (`lCBOpt-iso`, $s=-1$, sweeping the anti-collapse tolerance r) and three standard SAM (sweeping ρ), comparing sensitivity to each (see Appendix A for the choice of r). The reference $\bar\rho_{\mathrm{nom}}$ centring the standard-SAM sweep is the temporal mean of the learned radius $\rho^\star(\eta)$ (Eq. (23)) at $r=r_{\mathrm{nom}}$ (within 5% of its final-epoch value), so the run at $\rho=\bar\rho_{\mathrm{nom}}$ also checks the learned radius itself.

Table 11: SAM group (C). r_{mult}, ρ_{mult} are multiples of $r_{\mathrm{nom}}=0.50$ and $\bar\rho_{\mathrm{nom}}=2.2165$ respectively.

Run	Kind	Multiplier	Acc
S1	`lCBOpt-iso` (r)	$\times1.0$	0.8876
S2	`lCBOpt-iso` (r)	$\times0.95$	0.8865
S3	`lCBOpt-iso` (r)	$\times1.05$	0.8904
S4	standard SAM (ρ)	$\times1.0$	0.2168
S5	standard SAM (ρ)	$\times0.95$	0.1373
S6	standard SAM (ρ)	$\times1.05$	0.2399

Across the $\pm5\%$ sweep, `lCBOpt-iso` runs barely vary ($\Delta_{\mathrm{adapt}}=0.0039$, 0.8865–0.8904), whereas standard SAM swings widely ($\Delta_{\mathrm{SAM}}=0.1026$, 0.1373–0.2399), a 26.31× higher sensitivity. Here the uncapped target $\vartheta^\star = d/\|u\|^2$ scales with dimension ($d=44{,}426$) well above the curvature, so it saturates systematically at the anti-collapse upper bound $(h+\delta)/r$ rather than visiting the interior of $\mathcal{I}_{-1}$. Standard SAM, meanwhile, collapses (0.14–0.24) under $\rho\approx2.1$–2.3, far outside its stable regime [31]. A sensitivity study in SAM's own operating range is left open.

The level-set bridge (21) produces $\bar\rho_{\mathrm{nom}}=2.2165$, far outside the SAM literature's $\rho\approx0.01$–0.2 [31]. Because ϑ is clamped to the bound $(h+\delta)/r$ throughout training, $\bar\rho\approx\sqrt{r/(h+\delta)}$, and reaching SAM's bracket would need $r\approx10^{-5}$–10^{-3} rather than $r_{\mathrm{nom}}=0.50$, a regime §6.2 already showed unusable for $s=-1$: it sends $\vartheta\propto1/r\to\infty$, vanishing the update $u_s\propto r$ and freezing training. Matching the range without shifting r would instead need α extremely close to 1 ($\alpha\approx0.998$–0.99999), not the fixed $\alpha=e^{-1/2}\approx0.6065$. As flagged in §7.1, this mismatch may reflect either a limitation of the bridge itself or simply that ϑ sits at the cap boundary rather than at the interior fixed point Proposition 6.10 derives, and separating the two is left open.

Summary. Three of the five toy results are sanity checks: regularity and monotonicity (plus its repair) hold because ℓ is exactly quadratic there, and the SAM direction is collinear by construction for any ℓ. The two genuine results differ. The penalty coefficient's sensitivity moves $\vartheta^\star$ by 2.8–6.4× over a 16× range, justifying fixing $\zeta=d/2$ by the volume argument rather than by hand. The dimension sweep's ordering, LRD < BD < iso < diag, confirms the direction predicted by the K Picard solves of §7.2 but not the magnitude (2× measured against $K=5\times$ predicted), pointing to GPU kernel-launch overhead as a contributing factor.

On the real network the picture is more mixed. The benchmark shows genuine gains over fixed-ϑ `uCBOpt` on accuracy (0.912–0.915 against 0.909), NLL (0.252–0.257 against 0.272), and ECE (0.014–0.022 against 0.025), and beats IVON on NLL and ECE despite its 64-sample ensemble. The learned-precision variants are also tighter, across seeds, than both baselines in-domain, but wider than both on OOD. `uCBOpt-BD` is the most fragile variant, one of three seeds diverging at epoch 16, though this training-loss gap does not carry over to validation loss. The mechanism behind these gains, however, is weaker than the theory intends: Theorem 6.14's contraction condition fails by 10^7–10^9, and neither ϑ nor the adaptive SAM radius ever reach their theoretically intended values. What trains the network is instead the cap tracking curvature at $d=44{,}426$ parameters, numerically close to `uCBOpt-adapt`'s hand-tuned γc_t envelope [1], extended here across all four precision structures rather than only the isotropic case Singh et al. [1] cover.

8 Discussion and Conclusion

8.1 Discussion

The five points below revisit, with more critical distance than the results sections allowed, what the reproduction, the landscape diagnostic, and the spread-learning construction, together with its SAM application, actually establish, and where each departs from its own theoretical guarantees.

Lessons from the landscape diagnostic. Section 5's central finding is that every large gap between a candidate and $g^\star_{\max}$ is driven by family misspecification rather than by the choice of divergence (§5.3), and it is this finding that justifies spending the rest of the report on the candidate family rather than on a softer $D_{\max}$. But Case 4 already exposes the limits of this approach: no single Gaussian candidate can be simultaneously mode-seeking and support-covering against a bimodal target, and this is not a defect that a better precision repairs, since it concerns the location, not the spread, of a unimodal family. Every construction in Sections 6–7 remains, at its core, a single Gaussian, richer in precision but never in the number of modes. The possibilistic union $g_A \vee g_B$ mentioned in §5.1 is the natural next step.

Structural sign asymmetry. Theorem 6.20 gives an unconditional safeguard for $s = +1$, whereas Theorem 6.21 gives one only under the feasibility condition (17). At LeNet scale, that condition fails on 100% of steps for `lCBOpt-LRD` and on roughly one third of blocks for `lCBOpt-BD` at any given step (§7.3). This is not merely a symptom of the curvature proxy used throughout: the asymmetry is already present in the exact statements themselves. Theorem 6.20 bounds $\lambda_{\max}(S)$ by a constant independent of the current Newton step, while Theorem 6.21 bounds $\lambda_{\min}(S)$ by a quantity that shrinks as $\|u_s^{\mathrm{LRD}}\|^2$ grows. A better curvature estimate would shift M and m, but would not remove this dependence: the asymmetry is a property of the lower-CBO objective under this precision family, not an artefact of how curvature happens to be estimated. The effective-sign flip (Remark 6.22) is a reasonable response to this structural fact, but it also means that `lCBOpt-BD` and `lCBOpt-LRD` spend most, or a substantial part, of their steps running the $s = +1$ objective instead, which makes `lCBOpt-BD` closer to a mostly-$s = +1$ hybrid in practice than to a genuinely distinct method.

A chain of results, largely eclipsed by the caps. Section 6's contribution lies less in any single result than in the chain of implications connecting them. The raw CBO cannot identify a dispersion (Lemma 6.3, strict monotonicity) → this motivates a geometric penalisation via the level-set log-volume (Theorem 6.5) → stationarity then branches by geometry, for isotropic and diagonal precisions yielding an implicit fixed-point condition (Proposition 6.10) → solved by a damped Picard step under anti-collapse and positivity caps ((14), Theorem 6.14) → Algorithms 1 and 2 (`CBOpt-iso` and `CBOpt-diag`), while for general matrices, stationarity instead yields a rank-one, singular precision (Proposition 6.18) → repaired by LRD regularisation (16) → sign-asymmetric positivity safeguards and their block extensions (Theorems 6.20, 6.21, 6.24) → Algorithms 3 and 4 (`CBOpt-LRD` and `CBOpt-BD`).

However, as reported in **Geometry and safeguards** for $s = +1$ and in **SAM application** for $s = -1$ (§7.3), the isotropic and diagonal precisions this chain was built to characterise turn out never to be what actually operates at network scale. For $s = +1$, $H_s \succ 0$ before capping holds on 0.0% of steps for both `uCBOpt-iso` and `uCBOpt-diag`, so the precision sits at the positivity boundary ($(1 - \epsilon_{\mathrm{rel}}) \min_k(h_k + \delta)$ for `uCBOpt-iso` and $(1 - \epsilon_{\mathrm{rel}})(h_j + \delta)$ coordinatewise for `uCBOpt-diag`) of $\mathcal{C}_{+1}$ for the entire run. For $s = -1$, uncapped positivity holds on 100% of steps while $\vartheta^\star \approx d/\|u\|^2 \gg (h_j + \delta)/r$, so the iterate in `lCBOpt-iso` instead saturates the opposite, anti-collapse bound $(h_j + \delta)/r$ of $\mathcal{C}_{-1}$. In neither regime does the precision ever reach the interior stationary point of Proposition 6.10, and the cap, meant only as a guard, becomes the entire mechanism in practice. For $s = +1$, this is compounded by the fact that the learned value matches `uCBOpt-adapt`'s hand-designed envelope γc_t [1] up to construction and smoothing, with near-identical numbers on every metric (Acc 0.912 vs 0.910, NLL 0.257 vs 0.256), even though our construction covers all four precision structures where `uCBOpt-adapt` only covers the isotropic case. Whether an interior point ever becomes reachable, rather than perpetually capped, on a network where curvature varies less sharply across training is left open.

Unreliability of the learned SAM radius. The exact identity $\underline{\mathrm{CBO}}(g_{\eta,\rho}) = -\mathcal{E}_{\mathrm{SAM}}(\eta;\rho,0)$ (Proposition 7.1) holds only for the indicator candidate. Learning the SAM radius operationally faces two distinct difficulties: moving to a Gaussian candidate, necessary since Section 6's machinery learns precision for that family, already relaxes the identity into an L^2 approximation (Proposition 7.2), and the induced radius $\rho_\alpha = \vartheta^{-1/2}$ inherits the boundary saturation detailed above, ϑ sitting at the anti-collapse bound rather than at the stationary point. The measured $\bar{\rho}_{\mathrm{nom}} \approx 2.2$, an order of magnitude above the SAM literature's typical $\rho \in [0.01, 0.2]$ [31], cannot therefore be conclusively attributed to a flaw in the level-set bridge (21) itself. Separating the two causes would require evaluating settings where ϑ settles in the interior of $\mathcal{I}_{-1}$, left open. The reported insensitivity of the learned radius to r ($\Delta_{\mathrm{adapt}} = 0.0039$ for `lCBOpt-iso` against standard SAM's $\Delta_{\mathrm{SAM}} = 0.1026$) should thus be read as a comparison of two optimisers in suboptimal regimes, not as definitive evidence that the framework cannot learn a well-calibrated SAM radius at network scale.

Efficiency and out-of-distribution variance against IVON. A central motivation for the possibilistic approach is that inference relies on a single deterministic forward pass, avoiding IVON's 64-sample Monte-Carlo ensemble (Remark 4.2). `uCBOpt-diag` matches or exceeds IVON on accuracy and NLL, and `uCBOpt-BD` on ECE, at 64× lower inference cost (Table 10). On out-of-distribution detection, IVON leads all learned-precision variants, but what stands out is the pattern in variance: every method that learns a precision, IVON and all three possibilistic variants (`uCBOpt-iso`, `uCBOpt-diag`, and `uCBOpt-BD`), exhibits wider seed-to-seed dispersion on OOD metrics than fixed-ϑ `uCBOpt`, despite tighter or comparable spread in-domain (§7.3, **Benchmark results**). All four still outperform AdamW on OOD, confirming that learning a precision is not the issue. This reflects the standard underspecification phenomenon [34], where different solutions perform identically in-domain but extrapolate with higher variability on OOD inputs.

8.2 Conclusion

This report addresses a limitation of [1]: while mode updates are derived variationally, the precision of the Gaussian candidate is never derived from the variational objective itself, remaining fixed by hand or tracked via heuristics. After laying out the theoretical foundations (Section 3), reproducing a lighter variant of the reference benchmark (Section 4), and confirming on tractable posteriors that approximation gaps come from candidate misspecification rather than the choice of divergence (Section 5), Section 6 resolves this gap through a deductive progression (see **A chain of results** in §8.1). To resolve the collapse of unpenalised CBO objectives (Lemma 6.3), it introduces an extremal-volume penalty (Theorem 6.5). For isotropic and diagonal candidates, the resulting stationarity conditions yield fixed points solved via damped Picard iterations with anti-collapse and positivity caps (Proposition 6.10, Theorem 6.14). For full and block-diagonal matrices, the singular rank-one target (Proposition 6.18) is regularised into a low-rank-plus-diagonal form with sign-asymmetric positive-definiteness guarantees (Theorems 6.20–6.24). Section 7 translates these mechanisms into practical optimisers with linear per-step inversion complexity (Proposition 7.3), while connecting the lower CBO to SAM (Proposition 7.1) to derive an adaptive perturbation radius directly from the learned spread.

Empirically, our experiments evaluate both contributions across an exact quadratic toy and LeNet-5/Fashion-MNIST. On the toy problem, closed-form derivations confirm the penalty repair, validate the linear complexity hierarchy (LRD < BD < iso < diag), and highlight precision sensitivity to the penalty scale, justifying fixing $\zeta = d/2$. On the real network, `uCBOpt-diag` and `uCBOpt-BD` lower ECE to 0.020 and 0.014 (against 0.025 for fixed `uCBOpt` and 0.019 for 64-sample IVON) from a single deterministic forward pass, while improving accuracy to 0.915 (against 0.909 and 0.910) and NLL to 0.252 (against 0.272 and 0.258). In-domain variance across seeds is tighter for the learned-precision variants than for fixed `uCBOpt`, though this stability does not carry over to out-of-distribution evaluation, where IVON retains an advantage and seed-to-seed dispersion widens for all methods learning dispersion. In the SAM setting, the first-order linearisation gap scales empirically near ρ^2, and under a ±5% sweep of its tolerance parameter, the induced radius in `lCBOpt-iso` exhibits numerical insensitivity ($\Delta_{\mathrm{adapt}} = 0.0039$), whereas standard SAM destabilises ($\Delta_{\mathrm{SAM}} = 0.1026$).

These gains, however, come with four decisive operational limitations documented in our diagnostics. First, Picard contraction bounds fail at network scale, with local derivative estimates exceeding unity by seven to eight orders of magnitude ($L_\Phi \sim 10^7$–10^8). Second, the operational precision never settles at the theoretical interior fixed point, collapsing instead onto the anti-collapse and positivity cap boundaries (for $s = +1$, tracking curvature analogously to the heuristic envelope of `uCBOpt-adapt`). Third, the global $s = -1$ safeguard in `lCBOpt-LRD` is infeasible at every step on LeNet-5 (100% sign flips), an obstruction repaired by block-diagonalisation in `lCBOpt-BD`, which restores feasibility to 65.5% of active blocks, though at the cost of numerical fragility (diverging on one seed in `uCBOpt-BD`). Finally, because ϑ saturates the anti-collapse bound, the sensitivity study compares both optimisers in suboptimal operating regimes, and the induced SAM radius settles around $\bar{\rho}_{\text{nom}} \approx 2.2$, an order of magnitude above literature standards, leaving it open (see below) whether this discrepancy stems from boundary capping or from the Gaussian relaxation itself.

8.3 Open questions

The main points left open in this report are collected here for reference in their order of appearance.

- **Multimodal and non-parametric candidates.** Case 4 (§5.1) shows no unimodal Gaussian candidate can be both mode-seeking and support-covering against a bimodal target. Extending the construction of Section 6 to a possibilistic mixture $g_A \vee g_B$, learning each component's mode and precision separately, or using a non-parametric possibility candidate instead, are both left open (§5.1).
- **Varying the divergence order p during optimisation.** While §5.3 evaluates D_p only for fixed values of p, annealing p across training, starting with a soft divergence and hardening towards $D_{\max}$ as the candidate converges, was proposed but left for future implementation (Remark 5.1).
- **Theoretical guarantees of the volume penalty beyond the isotropic quadratic case.** Proposition 6.7 establishes an interior global extremum by leveraging the quadratic structure of ℓ on $\mathbb{R}^d$, which forces $u_s(\vartheta)$ to diverge at the spectral boundary (Lemma 6.6). Whether this interior extremum is preserved for general non-quadratic losses remains an open question (Remark 6.8). Similarly, extending this monotonicity-breaking property to diagonal, LRD, and BD parametrisations is left for future work, as their domains introduce cross-term couplings not covered by the scalar argument (§6.1).
- **Optimality of the LRD regularisation for rank deficiency.** Proposition 6.18 reveals that exact matrix stationarity yields a singular rank-one precision, regularised by the additive diagonal damping $S^{-1} = u_s^{\text{LRD}}(u_s^{\text{LRD}})^\top + \psi I$. While this mirrors low-rank covariance approximations in VI [29], whether this specific form represents the optimal regularisation in our context remains an open question (§6.3).
- **Joint recursion convergence and interior precision reachability.** Theorem 6.14 proves contraction only for ϑ at fixed η, leaving both simultaneous optimisation and the convergence of the joint (η, ϑ) recursion open (Remark 6.17). In practice at network scale, ϑ sits at the safeguard boundary throughout training rather than reaching the interior stationary point of Proposition 6.10, leaving open whether an architecture with smoother curvature variations could restore interior convergence (§8.1).
- **Theoretical foundations and empirical evaluation of the learned SAM radius.** Deriving the adaptive radius relies on a Gaussian relaxation of the exact indicator candidate (Proposition 7.2), leaving open whether this relaxation is theoretically sound to recover SAM's original ρ (§7.1). Furthermore, because ϑ saturates the anti-collapse bound, the learned radius settles around $\bar{\rho}_{\text{nom}} \approx 2.2$, an order of magnitude above literature values [31]. Determining whether this discrepancy comes from boundary capping or from the relaxation itself, alongside conducting an in-regime sensitivity sweep of standard SAM, remains for future investigation (§7.3, §8.1).

A Additional Experimental Detail and Code

This appendix collects the detail needed for reproduction not already given in §7.3: the optimiser settings for the learned-precision runs, the `uCBOpt-BD` divergence diagnosis, the positive-definiteness check, and the non-emptiness of the capping interval of `lCBOpt-iso` in the SAM experiments. Architecture, data, and training protocol are as in §4 (§4's Setup), which also gives the AdamW, IVON, and fixed-ϑ `uCBOpt` hyperparameters. The toy problem, the run perimeter, and the seed counts are stated in §7.3 and not repeated. **Repository links** to all experimental codes are provided at the end.

Optimisers. All learned-precision variants share $\beta_1 = 0.9$, $\beta_2 = 0.99999$, and prior precision $\delta = 2 \times 10^{-3}$. `CBOpt-iso` and `CBOpt-diag` solve the implicit fixed point with $K = 5$ damped Picard steps at damping $w = 0.5$, while `CBOpt-LRD` and `CBOpt-BD` compute ψ in closed form.

- **Benchmark** ($s = +1$). `uCBOpt-iso` and `uCBOpt-diag` with learning rate 10^{-2}, $h_0 = 0.05$, relative cap margin $\epsilon_{\text{rel}} = 0.2$, anti-collapse divisor $r = 10^{-2}$. `uCBOpt-BD` with learning rate 10^{-2}, $h_0 = 0.05$, additive safeguard margin $a = 1$, one block per parameter tensor.
- **Geometry check** ($s = -1$). `lCBOpt-LRD` and `lCBOpt-BD` with learning rate 2×10^{-3}, $h_0 = 0.1$, $a = 1$, feasibility margin $\xi = 0.1$, safeguard scale $\varphi = 2$.
- **SAM.** The three possibilistic runs are `lCBOpt-iso` ($s = -1$) with learning rate 2×10^{-3}, $h_0 = 0.1$, $\epsilon_{\text{rel}} = 0.2$, and $r \in \{0.500, 0.475, 0.525\}$. The three standard-SAM runs use SGD with learning rate 5×10^{-3}, momentum 0.9, weight decay 10^{-5}, and $\rho \in \{1, 0.95, 1.05\} \times \bar{\rho}_{\text{nom}}$.

`uCBOpt-BD` divergence (seed 2). One of the three `uCBOpt-BD` benchmark seeds diverged at epoch 16 (training loss $0.47 \to 1.9 \times 10^7$, reaching 9×10^7 by epoch 20, no recovery) and is excluded from every aggregate in Table 10 and Figure 5. By the Sherman–Morrison structure of H_s (Proposition 7.3), the diagonal part $P_\ell = \operatorname{diag}(h_\ell + \delta) - \frac{1}{\psi_\ell} I_{d_\ell}$ has smallest entry $m_\ell - 1/\psi_\ell \approx a\, m_\ell^2$ under $\psi_\ell = 1/m_\ell + a$ (Theorem 6.24(a)). With $m_\ell \approx 0.035$ on non-diverging blocks, this amplifies ill-conditioned directions by $1/(a\, m_\ell^2) \approx 800$, identical across seeds. We deliberately avoid step-norm clipping or damping [35]: the rank-one LRD regularisation already certifies $H_s \succ 0$ analytically, and adding heuristic damping would distort the variationally derived precision. Tracking the fraction of blocks with $\|u_\ell\|^2 \geq 1/M_\ell$ (§6.3) shows it jumping $0.52 \to 0.82 \to 1.00$ over epochs 15–17 on seed 2 (versus ≈ 0.52–0.67 on seeds 0 and 1), pointing to a stochastic gradient spike on that run rather than an intrinsic algebraic flaw.

Positive-definiteness check. For `CBOpt-iso` and `CBOpt-diag`, H_s is diagonal, so $H_s \succ 0$ reduces to checking, coordinatewise, the sign of $s(h_j + \delta - \vartheta)$ at the uncapped target $\vartheta^\star = d/\|u\|^2$ (isotropic) or $\vartheta_j^\star = 1/(u_j)^2$ (diagonal). For `CBOpt-LRD` and `CBOpt-BD`, what is tracked is the feasibility threshold $\|u_\ell\|^2 \geq 1/M_\ell$ (§6.3), which governs the existence of a regulariser ψ_ℓ certifying $H_s \succ 0$ (Theorem 6.21). The run flips to $s_{\text{e}} = +1$ once $\|u_\ell\|^2$ exceeds the margin $(1 - \xi)/M_\ell < 1/M_\ell$ (Remark 6.22), behind the flip rates of Figure 6.

Non-emptiness of the capping interval of `lCBOpt-iso`. For `lCBOpt-iso` ($s = -1$), the invariant interval $\mathcal{I}_{-1} = [\max_j(h_j + \delta)(1 + \epsilon_{\text{rel}}), \max_j(h_j + \delta)/r]$ of Eq. (15) is non-empty for $r < 1/(1 + \epsilon_{\text{rel}})$, i.e. $r < 0.833$ at $\epsilon_{\text{rel}} = 0.2$. Centring the SAM sweep on $r_{\text{nom}} = 0.50$ keeps $r \in [0.475, 0.525]$ well inside this range, so the cap $\max_j(h_j + \delta)/r$ is the active bound throughout the sweep and ϑ depends strictly on r.

Compute. All experiments were run on Google Colab (T4 GPUs).

Code. Three public repositories accompany this report: https://github.com/mariadass-gabriel/MDV_reprod (Section 4), https://github.com/mariadass-gabriel/MDV_lands (Section 5), and https://github.com/mariadass-gabriel/MDV_spread (Section 7), each containing the code and logs behind every number reported.

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